

RMCA Product Offering Function Guide

User Guide

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1 Introduction to Product Offerings

A product offering is an offer made by an operator to customers. It is derived from a product specification and is normally realized by one or multiple services such as voice, SMS, or data. Product offerings and all related entities are configured in Revenue Management Configuration Application (RMCA).



2 Product Offerings

2.1 Configurations in Product Offerings

A product specification is associated with a product offering, which defines the product offering's behavior and characteristics.

For more information on configuration of product offering within the Business Application Portal GUI, see [Configure Product Offerings](#).

Versioning of Product Offerings

For product offerings new versions can be created when they are in a published state. A new version in draft state includes the following options:

Add the following relations:

- Activate on expiry
- Optional for

A new optional for relation can only be added if the last version already has at least one optional for relation.

Note: Activate on expiry and optional for are not possible to combine.

Add or edit characteristics.

Editing allows:

- Edit the selection of personalization on instance level
- Edit the selection to allow values to be transferred
- Edit the number of Required and Allowed values
- Add new single values
- Edit the default status of existing single values
- Add new, or extend existing, value ranges

Add the following price configurations:

- Service usage configurations
- Refill configurations
- Discount on one time charge configurations
- Aggregated charges configurations



- Recurrence configurations
- Product exchange configurations

Change the following action values in existing price configurations:

- Service usage configurations
- Refill configurations
- One time charge configurations
- Discount on one time charge configurations
- Aggregated charges configurations
- Promotions configurations
- Recurrence configurations

Price configurations that can be removed or added in a new version:

- Partner settlement of subtype service usage configurations
- Partner settlement of subtype recurrence configurations
- Partner settlement of subtype one time charge configurations
- Product exchange configurations
- Recurrence configurations
- Aggregated charges configurations
- Discount on one time charge configurations
- Refill configurations
- Service usage configurations

Note: The price configuration which is enriched in agreement specification with a overlapping validity cannot be removed in a new version.

Other configurations remain unchanged. In a new version, existing service usage, refill, and recurrence configurations can be duplicated, and then edited.

2.2 Relationships in Product Offerings

Product offerings can be related to other entities during configuration.



Table 1 Product Offering Relationships

Relation to	Description
Product specification	A product specification is connected to a product offering to enable the use of business logic and entities configured on the specific product specification.
Product offering	A product offering can be connected to a product offering as part of a bundle, where several product offerings are connected as one bundle product offering. The bundle product offering can, for example, be used to set a combined recurring fee for several product offerings. A product offering can also be connected to a product offering to create relationships for sharing, discount, promotion, or recurring purchase. — For sharing, one or more consumer product offerings are connected to the provider product offering. A provider product offering can have relations to multiple consumer product offerings as long as they are connected to the same sharing specification. For example, members within a community can be offered different rates or individual limits but still be part of the same community. — For discount on usage, the base product offering to be discounted is connected to the discount product offering. For discount on one time charge, product offerings through which the discount is to be offered as a promotion, and product offerings to which the discount is to be applied, are connected. For each product offering connected with a discount applies for relation, at least one specific price configuration to discount must be selected in the discount on one time charges section. — For promotion, the product offering to be activated as part of a reward, and originating product offerings through which customers are offered the promotion, are connected to the promotion rule product offering. — For recurring purchase, the product offering to be purchased as part of an external notification is connected. Add-on: A product offering can be offered as an add-on together with another product offering through an optional for relation. Each product offering can have several add-on product offerings. Optional for relation does not allow the source product offering to be added, as an add-on product offering, to the target product offering. Trial: A product offering can be offered as a trial together with another product offering through an activate on expiry relation. A trial product offering (source) does not allow activate on expiry relation to other trial product offerings (source). A standard product offering (target) does not allow activate on expiry relation to any other product offerings.
Tags	A product offering is connected to a tag to enable indirect referencing of the product through the tag on instance level.



2.3 Business Strategies in Product Offerings

The values for business strategies are defined on product offering level. Values for global conditions, defined on product specification level, can be edited and used in a product offering.

Business strategies, once configured, are evaluated from top row, to bottom row.

2.3.1 Service Usage in Product Offerings

A service usage strategy is used to define the price of using a product. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the linear price applied for each minute of voice call and which bucket or tag to deduct from.

At least one service usage configuration must be included for each usage service that is connected to the product offering. The service type is determined by the connected normalized event specification.

2.3.2 Refill in Product Offerings

A refill is used to add a value on a billing account bucket or product bucket. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the number of units to refill and which bucket to add the refill to, or which counter product bucket to accumulate towards.

At least one refill configuration must be included for each refill service connected to the product offering.

2.3.3 One Time Charges in Product Offerings

One time charges are used to deduct one time fees from one or more customer billing accounts. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the purchase fee to be deducted when the product is bought, or activation fee.

One time charges can be configured to be paid before activation of the service. Several buckets can be referenced in a one time charge configuration and the order in which they are used can be set and changed. For example, a product purchase fee can be configured to be paid before activation of the product, and debited from a real money billing account first, and then from a bonus money billing account when the first account is depleted.

At least one one time charge configuration must be included for each one time charge event connected to the product offering. If a subsequent strategy is configured at product specification level, values must be defined for it at product offering level. An initial strategy must be upfront if the subsequent one is configured as upfront.



2.3.4 Aggregated Charges in Product Offerings

An aggregated charges strategy is used to define the price or discount of units aggregated during a billing period. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the linear price applied for each minute of voice call and which bucket or tag to deduct from, or the amount or percentage to reduce an aggregated charge.

At least one aggregated charges configuration must be included for each periodic counter that the aggregated charges strategy calculates from.

2.3.5 Discount on Billing (Billing Time Discount) in Product Offering

Billing time discount refers to an unconditional discount that applies to the total invoice amount without being based on any specific conditions. Billing time discount product offering applies discounts at billing time rather than as part of a standard provisioning or product life cycle events. Billing time discount type of product offering can only be created using a similar billing time discount product specification. This type of product offering provides a mechanism to reduce one-time charges (fixed amounts or percentage-based) such as purchase or activation fees according to business rules defined in the connected product specification.

Billing time discount product offering includes a product offering PolicyRef without a corresponding product specification PolicyRef. The life cycle of billing time discount type of product offering remains similar to base, however the first traffic event activation is not required.

2.3.6 Discount on One Time Charges in Product Offerings

Discount on one time charge are used to give a fixed or relative discount on one time charge, such as purchase fees. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the amount or percentage to reduce an upfront one time charge by when the product is bought, or activated.

At least one discount on one time charge configuration must be included for each discount on one time charge strategy connected to the product offering. The discount on one time charge logic is only run ones per basket session.

2.3.7 Promotions in Product Offerings

Promotions are used to boost the use of a particular service, or to promote a new service or product. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the number of units to give as a reward and which bucket to add the units to, or which product to activate as a promotion.



At least one promotions configuration must be included for each promotions strategy connected to the product offering.

2.3.8 Recurrences in Product Offerings

A recurrence strategy is used to, for example, set a fee for a product in a specific time period. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the number of units to deduct from a billing account bucket as a monthly recurring fee, or which counter product bucket to reset to zero.

Regular recurrence rules can be configured to allow external clients to manually trigger recurrence outside of the regular schedule. They can also be configured to enable pre-execution of the rule. Pre-execution helps prevent signaling spikes in a live system by allowing recurrence to execute early, within a defined window of time prior to the scheduled next execution.

At least one recurrence configuration must be included for each recurrence event connected to the product offering.

Recurrences on active product offerings can be prioritized using recurrence priority list in the Business Application Portal GUI. For more information, see [RMCA Product Offering Function Guide](#).

2.3.9 Lifecycle in Product Offerings

A lifecycle strategy is used to respond to changes in the product lifecycle state. For example, taking actions to prolong unit validity lost during a period in halt state when the product transitions back to active.

A lifecycle configuration is optional to define on product offering level even if a lifecycle strategy has been configured on the product specification.

2.3.10 Service Triggered in Product Offerings

A service triggered strategy is used to configure actions triggered by service use. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the number of units to deduct when the product is activated by service use, and which bucket to deduct from.

At least one service triggered configuration must be included for each service triggered event connected to the product offering.

2.3.11 Product Exchange in Product Offerings

A product exchange strategy is used to define which product bucket units to transfer and which recurrences to align between the source product instance



and the target product instance at a product exchange. Depending on the business logic configured in the connected product specification, different values can be defined. For example, all account bucket units are transferred but only recurrences connected with a certain strategy tag are aligned.

At least one product exchange configuration must be included in each exchange product offering. All product exchange configurations must be added to the same product exchange strategy.

2.3.12 Partner Settlement in Product Offerings

A partner settlement strategy is used to define the compensation for the partners involvement in a partner involvement group. Depending on the business logic configured in the connected product specification, different values can be defined. For example, the fixed amount of the customer revenue, that a partner is entitled for the involvement.

At least one partner settlement configuration must be included in each partner settlement product offering. Only postpaid payment method is applicable.

2.4 Configure Product Offerings

Create and manage product offerings by using the Business Application Portal GUI.

For more information on tasks that can be performed, see the following sections:

- Create a product offering, see [Create a Product Offering](#) section.
- View a product offering, see [View an Entity](#) section.
- Modify a product offering, see [Modify an Entity](#) on page 13 section.
- To publish a product offering, see [Publish an Entity](#) on page 14 section.
- Create new version of a product offering, see [Create New Version of an Entity](#) on page 14 section.
- Duplicate a product offering, see [Duplicate an Entity](#) section.
- Suspend a product offering, see [Delete a Product Offering](#) on page 16 section.
- Unsuspend a product offering, see [Unsuspend a Product Offering](#) on page 17 section.
- Discontinue a product offering, see [Discontinue a Product Offering](#) on page 18 section.
- Reintroduce a product offering, see [Reintroduce a Product Offering](#) on page 19 section.



- Delete versions of a product offering, see [Delete Obsolete Versions of a Product Offering](#) on page 20 section.

2.4.1 Create a Product Offering

A product offering is created from the **Product Offerings** list view which is accessed through the **Product Offer Configuration** UI application in the **Business Application Portal** menu.

Note:

Prerequisites

- A product specification.

Steps

1. In the **Product Offer Configuration** UI application, click **Product Offerings**.
2. In the **Product Offerings** list view, click **Create new** in the top right-hand corner.
3. In the **Create Product Offering** pop-up, perform the following steps:
 - a. Select the type of offering in the **Details** section.
 - b. Click **Next**.
 - c. Select a product specification to create the product offering from.
 - d. Click **Create**.

Result: The detail view of a new product offering is displayed, showing the sections and fields that create a product offering.

Note: Depending on the product offering type and the connected product specification, different relationships and business strategies are available for configuration.

4. In the **Details** tab, perform the following steps:
 - a. Enter a **Name** for the product offering.
 - b. Enter a **Description** for the product offering.
 - c. Enter an **External ID** for the product offering.
 - d. To make the entity unavailable for commercial use, define a **Discontinue from** date using context menu.
 - e. Select if this product offering is **Only use in bundles**.



Result: The details of the product offering are defined.

5. In the **Loan Due date** tab, perform the following steps:
 - a. Enter the **Due Date** for the loan product offering.
 - b. Define the **End user notifications**.
 - Select and drag a warning notification and define a period of the time before due date the warning is to be sent.
 - Select and drag a notification on the **Due date**.
 - c. Define the **External system notifications**.
 - Select and drag a notification on the **Due date**.
 - Select and drag a notification on the **Product deactivation**.

Result: The Loan Due Date of the loan product offering are defined.

6. In the **Characteristics** tab, inherited characteristics can be customized. Click **Inherit** to inherit a characteristic to the product offering and perform the following steps:

The **Details** can be customized as follows:

- a. Select a characteristic to customize.
- b. Click **Inherit**.

A list view displayed with a selected local characteristics.
- c. Click on a local characteristics and perform the following steps to customize the characteristic values.

In the **Edit characteristic** wizard, the **Details** can be customized as follows:

 - d. Enter a **Name** for the characteristic.
 - e. Edit the auto-generated **External ID** of the characteristic.
 - f. Enter a **Description** for the characteristic.
 - g. Select the **Value type**.
 - h. Depending on the value type selected, select any associated **Unit type or Derived value type**.
 - i. Click **Next**.

The **Settings** can be customized as follows:



- j. Select whether to **Enable customization** of the characteristic and, if enabled, define the level of **Personalization on instance level** and if values are allowed to be transferred at product exchange.
- k. Select if personalized values are to be tagged to indicate that the **Data is sensitive**. If enabled, select which **Type of sensitive data** the values contain and if encrypted data is to be decrypted before sending it to other application components.
- l. Enter the **Number of required values** at instance level.
- m. If customization is enabled, enter the **Number of allowed values** at instance level.
- n. Click **Next**.

Create the **Values** as follows:

Note: The cardinality of the values required is displayed, which is determined by the details defined in the previous step.

- o. Define any **Single values** by entering **Name**, **Value**, and **Unit**.
- p. Edit or restore the **External ID**. Select if the value is to be used as default.
- q. **Copy** single string values from a global data item list by clicking **Copy** and selecting the **Group**, **List name**, and **Values** to copy.
- r. Add **Referenced values** from a full global data item list by clicking **Add** and selecting the **Group** and **List name** to reference.
- s. Define any **Value range** by entering **Name**, **Min value**, **Max value**, and **Unit**. Edit or restore the **External ID**. Or select to **Use full range** of the selected value type.
- t. Click **Done**.

Result: The characteristic is customized and shown under the **Characteristics** tab.

- 7. In the **Life Cycle** tab, perform the following steps:
 - a. In the **Activation** section, define when the product activates.
 - b. If the product is activated by the **First traffic event**, define when the activation availability period starts and ends.
 - c. In the **Expiration** section, define when the product expires.
 - d. Select if extension of product validity on resumption from halt is to the end of the day on the new expiry date.
 - e. In **End-user notifications** section, select and drag a notification to send.



Result: The product life cycle is defined and shown under the **Life Cycle** tab.

8. In the **Relations** tab, do the following:
 - a. In **Product Offerings** tab, select the type of relation and click **Next**.
 - b. Select and drag one or more product offerings to connect with the selected relation type.
 - c. Click **Save**.

Result: One or more product offerings are selected and shown under the **Product Offerings** tab.

9. In the **Business Strategies** tab, do the following:
 - a. Select the tab of the type of business strategy to edit: Service Usage, Refill, Recurrences, One Time Charge, Aggregated Charges, Promotions, Life Cycle, Service Triggered, Scenario Triggered.

Note: Depending on the selected business strategy, different options are available for configuration.
 - b. Select an existing strategy from library.
 - c. Click **Create price configuration** to create a new price configuration.
 - d. Enter an **External ID** for the service usage price configuration.
 - e. Select the **Payment Context**.
 - f. If applicable, select the payment to be upfront.
 - g. If applicable, select a **Partner group ID**.
 - h. If applicable, select whether manual triggering of the rule is allowed or not.
 - i. If applicable, select whether to enable or not to enable pre-execution of the rule.
 - j. For strategies triggered by a recurrence-shifted schedule, select whether to use recurrence shift time or not. If not used, the time at which recurrence executes is set to midnight.
 - k. In the configuration tab, perform the following:
 - i. Click plus icon to define the conditional parameters.
 - ii. Click pencil icon to define the action values.
 - l. In the **External Id** tab, edit or restore the auto-generated external Id of the applicable row(s), action(s), or attribute(s).



m. In the **Personalization** tab, select the applicable row (s) and click **Enable personalization**.

n. Click **Done**.

Result: A price configuration is created for a business strategy and shown under the respective strategy tab.

10. Click **Save** in the top right-hand corner of the product offering detail view.

Result: By default, a product offering is saved in definition state and can be edited or deleted. To publish the product offering, click **Publish**.

2.4.2 View an Entity

Entities are viewed from their respective list views, which are accessed through the UI applications in the **Business Application Portal** dashboard.

Steps

1. In the UI application, click the tab of the type of entity to view.

Result: The entity list view is displayed, showing a list of existing entities.

2. Click an entity.

Result: The selected entity is displayed.

Note: An entity in **Published** state is viewed as read-only, while an entity in **Draft** state can be modified.

2.4.3 Modify an Entity

Entities in draft and published state are modified from their respective list views, which are accessed through the **Product Offer Configuration** UI application in the **Business Application Portal** menu.

Steps

1. In the **Product Offer Configuration** UI application, click **Product Offerings**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. To modify an entity, do any of the following:

a. Click the context menu option present at the right side of the entity row to expand it.



-Or-

- b. Click the entity to open the **Details** tab view and click the context menu in it.

Result: The selected entity is displayed.

3. Click **Offering details** in the context menu list.

Result: The **Edit offering details** dialog is displayed.

4. Modify any of the editable fields of the entity in the **Edit offering details** dialog.
5. Click **Save**.

Result: The modifications of the entity are saved, and the changes take effect immediately to all the versions of the specification.

2.4.4

Publish an Entity

Entities in draft state are published from their respective list views, which are accessed through the UI applications in the **Business Application Portal** dashboard.

Note: Publishing an entity also publishes all connected entities in draft state.

Steps

1. In the UI application, click the tab of the type of entity to publish.

Result: The entity list view is displayed, showing a list of existing entities.

2. Click an entity that is in **Draft** state.

Result: The selected entity is displayed.

3. Click **Publish** in the top right-hand corner of the page.

Result: A publish confirmation dialog is displayed.

4. Click **Publish**.

Result: The entity is successfully moved to **Published** state and is viewed as read-only.

2.4.5

Create New Version of an Entity



Steps

1. In the **Product Offer Configuration** UI application, click **Product Offerings**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. Select an entity that is in **Published** state, and do any of the following:
 - a. Click the context menu option present at the right side of the entity row to expand it.

Result: The context menu list is displayed.

-Or-

- b. Click the entity.

Result: The selected entity is displayed

3. Click **New Version** in the context menu list.

-Or-

Click **New Version** in the top right-hand corner of the page.

Result: A new version of the entity is created. The new version is displayed in **Draft** state and some of the tabs and fields can be modified.

To discard the new version, click **Revert Changes**.

A confirmation dialog is displayed.

In the confirmation dialog, click **Revert Changes**.

The new version is discarded.

2.4.6

Duplicate an Entity

Entities in published state are duplicated from their respective list views, which are accessed through the UI applications in the **Business Application Portal** dashboard.

Steps

1. In the UI application, click the tab of the type of entity to duplicate.

Result: The entity list view is displayed, showing a list of existing entities.

2. Click an entity that is in **Published** state.



Result: The selected entity is displayed as read only.

3. Click **Duplicate** in the top right-hand corner of the page.

Result: The entity is duplicated. The new duplicate is displayed in **Draft** state and can be modified.

4. Click **Save** to save the duplicate.

Result: By default, the duplicate is successfully saved in **Draft** state and can be modified.

Note: In some views, it is possible to discard duplication of an entity, as long as it is not saved.

To discard, click **Discard**.

A confirmation dialog is displayed.

In the confirmation dialog, click **Yes**.

The duplicated entity is discarded.

2.4.7

Delete a Product Offering

Entities only in published state are suspended from their respective list views, which are accessed through the **Product Offer Configuration** UI application in the **Business Application Portal** menu.

Steps

1. In the **Product Offer Configuration** UI application, click **Product Offerings**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. To delete a product offering, do any of the following:

- a. Click the context menu option present at the right side of the **Published** entity row to expand it.

-Or-

- b. Click an entity that is in **Published** state to open the **Details** tab view and click the context menu in it.

Result: The context menu list is displayed.

Note: The **Delete offering** option is enabled when the product offering is being used by a tag specification or NPG.



3. Click **Delete offering** in the list.

Result: A delete confirmation dialog is displayed.

4. Select the date on the confirmation dialog display, If required.

Note: If a date is selected to define the grace period, the product offering will be scheduled to get suspended on the selected date and it will be permanently deleted from the system after the grace period is over.

If a date is not selected and a grace period is not defined, the product offering is suspended. The suspended product offering is grayed out in the list view and displayed as Suspended.

5. Select the confirmation check box, and click **Delete Product Offering**.

Result: If grace period is defined, the product offering gets scheduled for suspension. Product offering will get deleted after the grace period is over. A yellow warning icon with the product offering is displayed in the list view. Hover over the icon to view suspension date.

If grace period is not defined, the product offering gets suspended and a message stating that the visibility of the product offering for other systems has been suspended is displayed. The suspended entity is grayed out in the entity list view, and displayed as Suspended.

2.4.8 Unsuspend a Product Offering

The suspended product offerings are unsuspended from their respective list views, which are accessed through the **Product Offer Configuration UI** application in the **Business Application Portal** menu.

Steps

1. In the **Product Offer Configuration UI** application, click **Product Offerings**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. Refine the list for suspended entities using the **Filter** option. To filter, type the relevant keyword in the **Filter** text box, or use **More filters** to select the **Suspended** state.

Result: The suspended entities are filtered in the list view.

3. To unsuspend a product offering, click the suspended entity to open the **Details** tab view and click the context menu in it.

Result: The context menu list is displayed.



4. Click **Unsuspend offering**.

Result: A confirmation dialog to unsuspend the product offering is displayed.

5. Click **Unsuspend Product Offering**.

Result: A message stating that the product offering has been unsuspended is displayed.

2.4.9 Discontinue a Product Offering

Entities only in published state are discontinued from their respective list views, which are accessed through the **Product Offer Configuration** UI application in the **Business Application Portal** menu.

Steps

1. In the **Product Offer Configuration** UI application, click **Product Offerings**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. To open the context menu list, do any of the following:

- a. Click the context menu option present at the right side of the **Published** entity row to expand it.

-Or-

- b. Click the entity that is in **Published** state to open the **Details** tab view, and click the context menu in it.

Result: The context menu list is displayed.

3. Click **Discontinue offering** in the list.

Result: A **Discontinue product offering** dialog is displayed in which a **Discontinue from** date can be set for the entity.

4. Define a date from which to discontinue the entity. Enter either a value manually in the **Discontinue from** field or click the calendar icon to select one from the data picker and click **Apply**.
5. Click **Save**.

Result: The discontinue date is saved and displayed in its own section above the entity details.

In the list view, discontinued entities are grayed out and marked as **Discontinued**. Click the discontinued entities in the list view, to view the discontinue date and time. Entities pending discontinuation are indicated by



a yellow warning icon. Hover over the warning icon to view discontinue date and time.

6. To change the discontinue date, click **Edit discontinue date** in the context menu and modify it.

Note: The **Edit discontinue date** is visible only when the product offering is in **Published** state and it is scheduled for discontinuity.

7. To remove the added discontinue date, click **Remove discontinue date** in the context menu.

Result: A confirmation dialog to remove the discontinue date is displayed. Click **Yes** to remove the date.

2.4.10

Reintroduce a Product Offering

The discontinued product offerings are reintroduced from their respective list views, which are accessed through the **Product Offer Configuration UI** application in the **Business Application Portal** menu.

Steps

1. In the **Product Offer Configuration UI** application, click **Product Offerings**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. Refine the list for discontinued entities using the **Filter** option. To filter, type the relevant keyword in the **Filter** text box, or use **More filters** to select the **Discontinued** state.

Result: The discontinued entities are filtered in the list view.

3. To reintroduce a product offering, click the discontinued entity to open the **Details** tab view, and click the context menu in it.

Result: The context menu list is displayed.

4. Click **Reintroduce offering**.

Result: A confirmation dialog to remove the discontinue date is displayed.

5. Click **Yes**.

Result: The product offering will be available for commercial use.



2.4.11 Delete Obsolete Versions of a Product Offering

Obsolete versions are the inactive/outdated product offering versions present in the test and/or production system. They are identified depending upon the state and validity of the product offering.

The PO version in use including the versions in draft state in the test and production systems, and unlocked and published versions in the test system, is not considered obsolete.

The PO versions of the future are considered obsolete only when they are overshadowed by another PO version and created prior to the PO version in use.

In the GUI, all published PO versions will be listed. The user can only select obsolete versions to delete. However, versions that are not obsolete will have their check boxes greyed out and cannot be selected.

In Test System, the version must be in a Locked state to be selectable for deletion.

Validity of the PO version

The PO versions can only be removed in the same order as they were created. For example, consider four PO versions: A, B, C, and D, where A is outdated, B is still active, and C and D are future PO versions. In this example, only version A is considered obsolete, while B is the current PO version, and C and D are locked future PO versions, making B, C, and D not obsolete.

Do the following steps to delete the obsolete versions in a product offering:

Steps

1. In the **Product Offer Configuration** UI application, click **Product Offerings**.

Result: The existing product offering entity list is displayed.

2. To delete the obsolete versions of a product offering, do any of the following:
 - a. Click the context menu option present at the right side of the entity row to expand it.

-Or-

 - b. Click the entity to open the **Details** tab view and click the context menu in it.

Result: The context menu list is displayed.

Note: The **Delete versions** option is not available if there are no obsolete versions present for the product offering.



3. Click **Delete versions**.

Result: The versions available for the product offering are displayed as a list.

- Note:**
- Only the obsolete versions are enabled such that the user can select them, and the other versions of the product offering are disabled and cannot be selected.
 - A warning message is displayed in the production system indicating the invalidation and disabling of the rollback point, if any valid rollback point exists for the business configuration and the obsolete PO within that configuration is removed.

4. Select the obsolete versions, and click **Delete** present in the top-right hand corner.

Result: A delete confirmation dialog is displayed.

- Note:** In the confirmation dialog, a warning message will be displayed in the production system indicating that if any of the selected versions have an active rollback point, this rollback point is invalidated, making it impossible to perform the rollback.

5. Select **Yes** in the confirm check box, and click **Delete versions**.

Result: The versions are deleted from the product offering.

2.5 Product Offerings Configuration

A product offering represents either a simple offering of a single product specification, or a bundle of one or more other product offerings.

In the **Create product offering** pop-up, select the product offering type to create: base, discount on usage, discount on billing, discount on one time charge, discount on aggregated charges, bundle, product exchange, loan, or partner settlement. Select a product specification from the pop-up. The pop-up displays product specifications in active state. To view product specifications in definition state, use the key combination Ctrl+Alt+A.

- Note:** When changes are made to a product specification in definition state that is connected to a product offering, do related changes in the product offering as well.

Entity List View

The entity list view presents all existing entities and the option to create a new one. Use the filter to search for existing entities. Narrow the search by selecting different variables. The specifications can be filtered by entity state, including discontinued status, suspended status and type.



In the list view, if a product offering is created from an external catalog using a product offering template, the **Created by** column shows External system and the **External Id** column shows the external Id of the template.

Product offering gets scheduled for suspension once grace period is defined while deleting the product offering. In the list view, a yellow warning icon with the product offering which is scheduled for suspension is displayed.

In the list view, discontinued/suspended entities are grayed out and marked as **Discontinued/Suspended**. Click the discontinued entities in the list view, to view the discontinue date and time. Entities pending discontinuation are indicated by a yellow warning icon. Hover over the warning icon to view discontinue date and time.

Click an offering displayed in the list view to edit, delete, or publish (activate) a draft, or to view, duplicate, or create a version of a published offering, or to suspend, unsuspend, discontinue, or to reintroduce a product offering.

Entity Detail View

A new entity is configured in the entity detail view which presents the configurable sections. Different actions, visible in the action bar, can be performed on the entity depending on its current state. The entity state is shown in the version section.

In the detailed view, if the product offering is created from an external catalog using a product offering template, it is denoted as External system under the **Details** section.

In a test system, the options to lock and deactivate an active entity are available.

Any validation errors received on attempting to save or activate an entity are highlighted in the detail view. A **Warnings** section shows which errors must be corrected to enable completion of the action.

Activating a product specification or any lower-level entity also activates all connected entities in definition state. Product offerings must be activated independently.

An active entity with a newer version in definition state, shows the different versions in the top right-hand corner of the detail view. Different entities permit different changes in new versions.

The functions displayed depend on the functional level of the system.

2.5.1 Details Tab in Product Offerings

Enter a name, description, and external ID. The external ID needs to be unique to address it correctly. The assigned ID is unique and cannot be edited.

If a product offering is created from an external catalog using a product offering template, the **Created by** shows External system.



The offering details can be modified through the context menu. To edit the entity fields, click **Offering details** in the context menu list.

The product offering can be suspended and discontinued through the context menu. Similarly, the suspended and discontinued entities are unsuspended and reintroduced through the context menu.

A discontinue date can be set through the context menu. If the entity is discontinued, the **Edit discontinue date** option is enabled in the context menu, and can be used to edit or remove the discontinue date.

If there exists any obsolete versions for the product offering, the **Delete versions** option is enabled. To delete, the user can select the obsolete versions and click **Delete versions**.

Optionally, restrict use of the offering to bundles only.

Version

Current entity validity period, state, creation date, last modification date, and related product specification are displayed. A note unique to each version of the offering can be added to describe, for example, the reason for creating the version.

Related product specification

The product specifications related to this product offering are displayed. Click the toggle icon to navigate into the respective **Relations** tab.

Used by

All entities that use the product offering are displayed.

Tags

All tags that are assigned to the entity are displayed.

2.5.2

Life Cycle Tab in Product Offerings

A product has a lifecycle during which it transitions through different states, for example from available to active to expired. End-user notifications can be sent at, and at defined offsets prior to, product lifecycle state transitions.

Define when the product activates and expires:

— Product activates on

Specify whether the active period starts on product instantiation, on a specified date, or on first traffic event. Activation by first traffic event is only



available when the product offering is connected to a sharing specification or an RFSS with provisioning system set to CPM.

For products activated by first traffic event, specify whether the product is available for activation from product instantiation or from a specified date. The availability period can be set to never expire, to expire after a specified duration, or to expire on a specified date.

— Product expires on

For all activation methods, the product can be set to never expire, to expire after a specified duration, or to expire on a specified date.

Select if the expiration date of a limited-validity product is extended to end of day on resuming from a halt state. By default, validity is extended by the exact length of time in halt but can optionally be extended further to the end of the day on the new expiration date.

End-user Notifications

Define the end-user notifications to send. Notifications can be sent at availability expiration, product activation, product expiration, and product termination. One or more reminders can be sent at various offsets prior to availability expiration, product expiration, and product termination.

Note: Product expiration notifications can be defined even if the product offering is configured without an expiration date. This is because the state transition dates can be personalized at instance level.

2.5.3 Relations Tab in Product Offerings

Connect other product offerings to the product offering.

In the Product Specification tab, the connected product specification is listed and can be viewed. Identification resources connected to the product specification are listed. If number portability is enabled on product specification level, it can be disabled

In the Product Offerings tab, select type of relation: **Optional for, Provides to**. The related product offering is to be offered as an add-on product to new product offering.

Connect other product offerings to the product offering for the following types of configuration:

Table 2 Product Offering to Product Offering Relations

Configuration	Relation Type	Description
Bundle	Bundles	Connect the member product offerings of a bundle and select the relation type. Relation between the main product offering and bundle product



Configuration	Relation Type	Description
		offering can be either strict or flexible.
Discount on Usage	Discount for	Connect the base product offering to be discounted.
Discount on One Time Charge	Discount applies for	Connect product offerings to which the discount is to be applied. For each product offering connected with a discount applies for relation, at least one of its upfront one time charge price configurations must be selected in the discount on one time charges section.
	Promotion for	Connect product offerings through which the discount is to be offered as a promotion.
Sharing	Provides to	Connect the consumer product offering to the provider product offering. To connect, the consumer product offering and the provider product offering must use the same sharing specification.
	Consumes from	Which provider product offering the consumer product offering consumes from is displayed.
Promotion	Promotion reward	Connect product offerings to be activated as part of a promotion.
	Promotion strategy for	Connect originating product offerings through which customers are offered the promotion.
Add-on	Optional for	Connect a product offering that is to be offered as an add-on product, to another product offering.
Trial Period	Activate on expiry	Connect a standard product offering to a trial product offering. The standard product offering is activated after the trial period expires.

After a product offering has been added for a relation type, use context menu to edit the list of product offerings which that specific relation type applies to.

Note: Activating the product offering activates all connected product offerings that are in state definition.

Relation Graph

This tab gives an overview of the product offering and its connected entities. A relation graph visualizes how the entities relate to each other.

The following actions are possible:

- Use the filter above the graph to search an entity by its name, type, and resource context where applicable. Click Remove filter to reset the filter.
- Click minus/plus at the top of the graph-field to collapse/expand the nodes along a vertical line.



- Click plus on the main node to view the connected nodes.
- Click minus on the main node to close the branch.
- Hover over or click a node for a focused view of the node and those connected to it. A click also displays the relation types.
- Click a node to view detailed information about the entity in the details view to the left. Use the context menu in the details view to go to the configuration page of the entity, remove the relation, change relation type, or define sharing scope.
- Click on the graph to drag it around in the window.
- Use the zoom slider at the bottom-right corner of the relation graph to adjust zoom.

2.5.4 Characteristics Tab in Product Specs

Characteristics can be inherited from the connected product specification, and where applicable, customized.

Clicking on an inherited characteristic displayed in the list view opens the **Edit characteristic** wizard. The wizard has three steps:

- **Step 1 - Details**
Edit the characteristic details.
- **Step 2 - Settings**
If customization is enabled, edit the level of personalization at instance level. Edit the number of required and allowed values.
- **Step 3 - Values**
Edit the characteristic values. Select which individual values or ranges to use from the original characteristic. New single values can be added if a value range is defined in the original characteristic. One or more single string values can be copied from a global data item list if the list was referenced in the original characteristic.

2.5.5 Business Strategies Tab in Product Offerings

Define business strategies for the entity.

- For information on business strategies of type **Refill**, see [RMCA Refill Function Guide](#).
- For information on business strategies of type **One Time Charge**, see [RMCA One Time Charge Function Guide](#).



- For information on business strategies of type **Discount on One Time Charges**, see RMCA Discount Product Function Guide.
- For information on business strategies of type **Promotion**, see RMCA Promotions Function Guide.
- For information on business strategies of type **Product Exchange**, see RMCA Exchange Product Function Guide.

Service Usage

Connect a service usage strategy created in the product specification and configure its values.

At least one service usage configuration must be included for each usage service connected to the product offering. The service type is determined by the connected normalized event specification.

For each usage service and price subtype it is possible to define one default price configuration and one for each price group. The default price configuration is the one used if no other price configuration is selected for the product. The price groups represent different commercial segments, for example student or corporate, and are fetched from global data item. The price groups enable a product to provide different prices depending on which commercial segment a customer belongs to. If a default or price group price configuration has been selected in a previous active version, then one must also be selected in the newer version. In a newer version the price configuration can be changed but the payment context must either correspond to the payment context in the previous version, or be generic. In the new version of product offering, it is permitted to delete unwanted and unused price configurations to keep configurations manageable as they grow with each new version. It is also possible to not select any price configuration as default. In the new version of product offering if it is required to delete the default price configuration, this can be achieved by first assigning another price configuration as the new default before deletion. If a price configuration is already been used in an agreement specification, It will not be deleted.

A service usage configuration includes dimensions and actions from the service usage strategy created on the product specification. For dimensions, use values created on product specification level, or create new ones in the configuration tab. Create values for actions in the configuration tab. At least one statement must be configured.

Service usage, once configured, is evaluated from top statement, to bottom statement.

The following options can be selected when setting values for the conditions. Depending on the strategy created in the product specification, different options from the table below are displayed.



Table 3 Service Usage Condition Values

Parameter Value	Description
Available Units	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If a referenced characteristic has no defined value, the condition is evaluated to false.
Bucket Balance	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If the referenced characteristic has no defined value, the condition is evaluated to false.
Community Membership	No options are available.
Dates	The specific dates, or list of dates, the price is valid.
Day of Week	The days of the week the price is valid.
Tiered Rating	The numerical value of the upper or lower limit. The value is included in the tiered interval.
Time and Day of Week	The days of the week and the times of day the price is valid.
Time of Day	The times of the day the price is valid.
Service Specific Parameter Values	Define the value to compare the parameter against. The service specific parameter can be compared against one of the following values: — A predefined single value in the same parameter. — A custom value within a range defined in the same parameter. — A global data item list. — A second common characteristic. Available options depend on the source selected in the condition, the parameters set in the normalized event specification, and the characteristics defined in the characteristic set specification. Note: When selecting a list name from global data item, start typing and the filtering of list names begins.
User Defined Parameter Values	Define a value to compare the common characteristic against. The selected common characteristic can be compared against one of the following values: — A predefined single value in the same characteristic. — A custom value within a range defined in the same characteristic. — A second common characteristic.



Parameter Value	Description
	Available options depend on the characteristics defined in the characteristic set specification.

The following options can be selected when setting values for the actions. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 4 Service Usage Action Values

Parameter Values	Description
Accumulate Consumed Units	Select the product bucket or bucket tag to accumulate the consumed units. The consumed units can be accumulated to a product bucket or bucket tag of type counter or periodic counter.
Apply Absolute Discount	Enter a value for the fixed amount that the linear price, session start fee, or session setup fee, is reduced by.
Apply Relative Discount	Enter a value for the relative amount in percent, that the linear price, session start fee, or session setup fee, is discounted.
Clear and Add to Product Bucket	Select a product bucket and enter the value to add after the bucket has been cleared of all existing values. Set the validity of the added units. If a start date is not defined, validity starts at service usage trigger time. If a duration or end date is not defined, the validity is unlimited. For monetary transactions the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.
Deduct Linear Price	Select the product bucket or bucket tag to deduct the price from, or billing account bucket is displayed. For a bucket tag, only referenced buckets defined with the selected price unit type are processed during evaluation. Note: Do not combine several linear price actions in the same strategy that directly address and also via a tag targets the same product bucket. Enter the price to deduct per consumed unit. Where applicable, select the product bucket or bucket tag to accumulate the cost. The cost can be accumulated to a product bucket or bucket tag of type counter or periodic counter. For monetary transactions the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can



Parameter Values	Description
	show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes.
Deduct Session Setup Fee	Select the product bucket or bucket tag to deduct the fee from, or billing account bucket is displayed. For a bucket tag, only referenced buckets defined with the selected price unit type are processed during evaluation. Enter the fee to deduct. Where applicable, select the product bucket or bucket tag to accumulate the cost. The cost can be accumulated to a product bucket or bucket tag of type counter or periodic counter. For monetary transactions the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes.
Deduct Session Start Fee	Select the product bucket or bucket tag to deduct the fee from, or billing account bucket is displayed. For a bucket tag, only referenced buckets defined with the selected price unit type are processed during evaluation. Enter the fee to deduct. Where applicable, select the product bucket or bucket tag to accumulate the cost. The cost can be accumulated to a product bucket or bucket tag of type counter or periodic counter. For monetary transactions the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define. Note: Values for this action may be overridden by external request parameters if enabled in the product specification. The override is only applied if the action is set to be applied.



Parameter Values	Description
Refund to Bucket	The amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.
Replace base price	Enter a value for the fixed amount that the linear price, session start fee, or session setup fee, is reduced by.
Send End-user Notification	Select the notification to send. A common description of the selected notification is displayed. The available values and common descriptions are fetched from NTF. The parameter value is optional to define.
Send External Notification	Select the action type and a product offering the action applies to. The action type determines the type of action to trigger. The available values are fetched from global data item. The parameter value is optional to define.
Set Charging Interval	Enter a value for the minimum number of units to charge if a service is used.
Set Product Bucket Value	Select a product bucket and enter the value to set the bucket to. The parameter value is optional to define.
Set Response to Network	Select a value defined on the normalized event specification, or if a value range is specified on the normalized event specification, enter a single value.

Aggregated Charges

Configure the values of the aggregated charges strategy, created on product specification level.

At least one price configuration must be included for each periodic counter that the aggregated charges strategy calculates from.

For each periodic counter, it is possible to define one default price configuration, one for each price reference, and one for each price group. Each price reference or price group must have a price configuration defined for all periodic counters.

— Default price configuration

The default price configuration is the one used if no other price configuration is selected for the product. A default price configuration is selected automatically.

— Price reference



The price references represent a combination of price configurations for the different periodic counters. The price references enable a product to provide different prices depending on the combination selected.

— Price group

The price groups represent different commercial segments, for example student or corporate, and are fetched from global data item. The price groups enable a product to provide different prices depending on which commercial segment a customer belongs to.

If a default price configuration, price reference, or price group has been defined in a previous active version, then one must also be selected in the newer version. In a newer version, the price configuration can be changed but the payment context must either correspond to the payment context in the previous version, or be generic. In the new version of product offering, it is permitted to delete unwanted and unused price configurations to keep configurations manageable as they grow with each new version. It is also possible to not select any price configuration as default. In the new version of product offering if it is required to delete the default price configuration, this can be achieved by first assigning another price configuration as the new default before deletion. If a price configuration is already been used in an agreement specification, It will not be deleted.

An aggregated charge configuration includes dimensions and actions from the aggregated charge strategy created on the product specification. For dimensions, use values created on product specification level, or create new ones in the configuration tab. Create values for actions in the configuration tab. At least one statement must be configured.

Aggregated charges, once configured, is evaluated from top statement, to bottom statement.

The following options can be selected when setting values for the conditions. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 5 Aggregated Charges Condition Values

Parameter Value	Description
Bucket Balance	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If the referenced characteristic has no defined value, the condition is evaluated to false.
User Defined Parameter Values	Define a value to compare the common characteristic against. The selected common characteristic can be compared against one of the following values: — A predefined single value in the same characteristic.



Parameter Value	Description
	— A custom value within a range defined in the same characteristic. — A second common characteristic. Available options depend on the characteristics defined in the characteristic set specification.
Tiered Rating	Define the numerical value of the upper and lower limit. The value is included in the tiered interval. The tiered interval can be selected to have no upper limit.

The following options can be selected when setting values for the actions. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 6 Aggregated Charges Action Values

Parameter Value	Description
Apply Absolute Discount	Enter a value for the fixed amount that the aggregated charges are reduced by. Select the product bucket to refund the fixed discount amount, or billing account bucket is displayed. For monetary transactions the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.
Apply Relative Discount	Enter a value for the relative amount in percent that the aggregated charges are discounted. Select the product bucket to refund the relative discount amount, or billing account bucket is displayed. For monetary transactions the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.
Deduct Linear Price	Select the product bucket or bucket tag to deduct the price from, or billing account bucket is displayed. For a bucket tag, only referenced buckets defined



Parameter Value	Description
	with the selected price unit type are processed during evaluation. Enter the price to deduct per consumed unit. Where applicable, select the product bucket or bucket tag to accumulate the cost. For monetary transactions the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.

Discount

A discount product offering contains all information required to apply a discount to a base product offering.

There are different types of discount product offerings: discount on usage, discount on billing, discount on one time charge, and discount on aggregated charges. A product offering of type discount on usage connects the base product offering to be discounted based on the usage. A product offering of type discount on billing applies discounts at billing time rather than as part of a standard provisioning or product life cycle events. A product offering of type discount on one time charge provides discounts based on items selected in a single order, or items selected and bought during a given time period, including previous orders. A product offering of type discount on aggregated charges applies discount on units aggregated during a billing period.

Discount on billing is configured using a tree based rule. Tree based rule consists of two parts: evaluation block and action block. Multiple evaluation block and action block joined by logical conditions forms a tree based rule. Evaluation block and action block can be configured using the **Business Application Portal GUI**. A product offering of type discount on billing applies discount at billing time on invoice amount by using either a relative (percentage) discount or fixed discount.

Table 7 Action Block

Type	Description
Relative Discount	Discount expressed in relative terms such as percentage (e.g., 5%). The relative discount UoMs supports all ratio related measures in global data items. e.g., percentage, permille or in basis points.
Fixed Discount	Discount expressed as a fixed amount (e.g., \$5). The fixed discount will be in any currency or UoM that is defined in measures e.g. USD, SEK or EURO.



Table 8 Billing Time Discount Parameters

Parameter Value	Description
Applicable on	Select the type of amount on which the billing time discount is applicable.
Discount on	Enter the amount on which the discount is applicable
Unit	Select the unit. Note: For relative discount, units are percentage, permille or in basis points etc. For fixed discount, units are monetary like USD etc.
Price Configuration Name	Enter the desired name for price configuration.
Payment context	Select the payment context as required. Note: For billing time discount, default payment context is postpaid.
Evaluation Scope	Select the Evaluation scope as invoice.

Table 9 Billing Time Discount Action Values

Actions	Description
Apply Fixed Discount	Use this action to apply fixed discount on the base amount. Example: Invoice amount is 50\$ and applied fixed discount is 5\$. Invoice amount post discount is 50\$-5\$ = 45\$.
Apply Relative Discount	Use this action to apply relative discount on the base amount. Example: Invoice amount is 50\$ and applied relative discount is 5%. Calculated discounted amount is 5% of 50\$ = 2.5\$. Invoice amount post discount is 50\$-2.5\$ = 47.5\$.

Life Cycle

Connect a life cycle strategy created in the product specification and configure its values.

At least one life cycle configuration must be included for each life cycle event connected to the product offering.

For each life cycle event it is possible to define one default price configuration and one for each price group. The default price configuration is the one used if no other price configuration is selected for the product. The price groups represent different commercial segments, for example student or corporate, and are fetched from global data item. The price groups enable a product to provide different prices depending on which commercial segment a customer belongs to. If a default or price group price configuration has been selected in a previous active version, then one must also be selected in the newer version. In a newer version the price configuration can be changed but the payment context must either correspond to the payment context in the previous version, or be generic. In the new version of product offering, it is permitted to delete unwanted and unused price configurations to keep configurations manageable as they grow



with each new version. It is also possible to not select any price configuration as default. In the new version of product offering if it is required to delete the default price configuration, this can be achieved by first assigning another price configuration as the new default before deletion. If a price configuration is already been used in an agreement specification, It will not be deleted.

A life cycle configuration includes dimensions and actions from the life cycle strategy created on the product specification. For dimensions, use values created on product specification level, or create new ones in the configuration tab. Create values for actions in the configuration tab. At least one statement must be configured.

A lifecycle strategy, once configured, is evaluated from top statement, to bottom statement.

The following options can be selected when setting values for the conditions in the lifecycle strategy configuration. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 10 Life Cycle Condition Values

Parameter Value	Description
Available Units	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If a referenced characteristic has no defined value, the condition is evaluated to false.
Bucket Balance	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If a referenced characteristic has no defined value, the condition is evaluated to false.
Reason for product resumption	Select a reason code. Available values for reason code are fetched from global data item.
Reason for product halt	Select a reason code. Available values for reason code are fetched from global data item.
State before product halt	Select the previous product state.
User Defined Parameter Values	Define a value to compare the common characteristic against. The selected common characteristic can be compared against one of the following values: — A predefined single value in the same characteristic. — A custom value within a range defined in the same characteristic. — A second common characteristic.



Parameter Value	Description
	Available options depend on the characteristics defined in the characteristic set specification.

Table 11 Life Cycle Action Values

Parameter Value	Description
Accumulate product resumption events	Select the product bucket or bucket tag to which to accumulate the events. The parameter value is optional to define.
Prolong unit validity by halt period	Select one or more product buckets for which to prolong the unit validity. Select if the extension should be to end of day on the new valid to date. The parameter value is optional to define.
Set Characteristic Value	Select a value from the list of defined values for the characteristic. Only single values are available to be set. The parameter value is optional to define.

Recurrences

Connect a recurrence strategy created in the product specification and configure its values. The schedule defined in the product specification determines when the rules are run.

For each recurrence event it is possible to define one default price configuration and one for each price group. The default price configuration is the one used if no other price configuration is selected for the product. The price groups represent different commercial segments, for example student or corporate, and are fetched from global data item. The price groups enable a product to provide different prices depending on which commercial segment a customer belongs to. If a default or price group price configuration has been selected in a previous active version, then one must also be selected in the newer version. In a newer version the price configuration can be changed but the payment context must either correspond to the payment context in the previous version, or be generic. In the new version of product offering, it is permitted to delete unwanted and unused price configurations to keep configurations manageable as they grow with each new version. It is also possible to not select any price configuration as default. In the new version of product offering if it is required to delete the default price configuration, this can be achieved by first assigning another price configuration as the new default before deletion. If a price configuration is already been used in an agreement specification, It will not be deleted.

A recurrence configuration includes dimensions and actions from the recurrence rules created on the product specification. For dimensions, use values created on product specification level, or create new ones in the configuration tab. Create values for actions in the configuration tab. At least one statement must be configured.

For regular recurrence rules triggered by a recurrence-shifted schedule, select if recurrence shift time is to be used in addition to the date. If not, the time at which recurrence executes is set to midnight (00:00:00.000). A recurrence-shifted schedule that repeats every day must always use recurrence shift time.



or regular recurrence strategies, select if external clients are allowed to trigger the recurrence manually outside scheduled execution dates. Additionally, select if pre-execution of the rule is to be enabled. Pre-execution helps prevent signaling spikes in a live system by allowing recurrence to execute early, within a defined window of time prior to the scheduled next execution.

Note: It is up to the external client to interpret the setting and only perform manual triggering when it is allowed according to the configuration.

For all types of recurrences except recurrence failure, optionally select a partner group ID. The partner group ID is used to correlate the customer price configuration with the partner involvements in a revenue sharing scenario. This is needed to know for which generated revenue an amount needs to be shared with partners, and to trigger the partner settlement process. At execution time the partner group ID of the customer price configuration is used to identify the partner involvements.

An option to edit price matrix external IDs is available when at least one row has been fully configured. In the price matrix external IDs pop-up, the auto-generated external IDs for the rows, actions, and attributes are displayed. The external IDs can be edited and restored to the auto-generated ones, and must be unique within its own level. The price matrix external IDs are optional to define.

The following actions and attributes can be edited:

- Add to Product Bucket
 - Value
 - Validity
 - Carry over
- Deduct from Bucket
 - Fee
- Set Product Bucket Value
 - Value

Recurrence, once configured, is evaluated from top statement, to bottom statement.

The following options can be selected when setting values for the conditions in the recurrence configuration. Depending on the strategy created in the product specification, different options from the table below are displayed.

The following options can be selected when setting values for the actions. Depending on the strategy created in the product specification, different options from the table below are displayed.



Table 12 Recurrence Condition Values

Parameter Value	Description
Available Units	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If a referenced characteristic has no defined value, the condition is evaluated to false.
Bucket Balance	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If a referenced characteristic has no defined value, the condition is evaluated to false.
Manually Triggered	Select if the parameter evaluates true when recurrence has been manually triggered (Yes) or when it has not (No).
Previous product state	Select the previous product state and optionally a reason code. Available values for reason code are fetched from global data item.
Reason for product state change	Select a reason code. Available values for reason code are fetched from global data item.
Reason for product resumption	Select a reason code. Available values for reason code are fetched from global data item.
Reason for product halt	Select a reason code. Available values for reason code are fetched from global data item.
State before product halt	Select the previous product state.
User Defined Parameter Values	Define a value to compare the common characteristic against. The selected common characteristic can be compared against one of the following values: — A predefined single value in the same characteristic. — A custom value within a range defined in the same characteristic. — A second common characteristic. Available options depend on the characteristics defined in the characteristic set specification.

The following options can be selected when setting values for the actions. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 13 Recurrence Action Values

Parameter Value	Description
Accumulate recurrence events	Select the product bucket or bucket tag to accumulate the number of recurrence events. The parameter value is optional to define.



Parameter Value	Description
Accumulate recurrence resumption events	Select the product bucket or bucket tag to which to accumulate the events. The parameter value is optional to define.
Add to Billing Account Bucket	Enter the value to add to the billing account bucket. Select whether or not to prorate the amount. The amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. P Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes.
Add to Product Bucket Note: It is only possible to add units to a product bucket of type Account with multiple value containers.	Select a product bucket and enter the value to add to it. Note: — If the carry over option is selected on product specification level, units can only be added to product buckets of type account. If units are to be added to a product bucket of type counter, the carry over option must be left unselected on product specification level. — Units cannot be added to product buckets of type account with a single value container. In positive and negative prorating, select whether or not to prorate the amount. Set the validity of the added units. If a start date is not defined, validity starts from the beginning of the day recurrence executes. If a duration, end date, or number of recurrence periods are not defined, the validity is unlimited. If validity is aligned to recurrence period with carry over, and the validity is greater than 1 recurrence period, define an absolute value or a percentage of the unspent units in each value container to be carried over at the end of the first period. If the carry over value is set to zero, no units are carried over into the second period. If units are to be carried over to a different product bucket, select which product bucket to carry over to. The selected product bucket must be of same type (account) and measure as the product bucket to which new units are added. Note: Positive prorating for reactivation always defines recurrence period validity without carry over. For monetary transactions, the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data



Parameter Value	Description
	item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes.
Change Product State	Select the status of the product, either suspended or inactivate. If suspended it is optional to define a validity time. When the validity expires, the product is inactivated. The validity must be greater than zero. If no validity is set, the product is suspended without any specific duration. The parameter value is optional to define.
Deduct from Bucket	Select the product bucket or bucket tag to deduct from, or billing account bucket is displayed. More than one bucket tag can be added. The bucket tags in the list are evaluated from top to bottom. Drag the product buckets or bucket tags to reorder them in the list. For a bucket tag, only referenced buckets defined with monetary units are processed during evaluation. Enter the monetary fee to deduct. Where applicable, select the product bucket or bucket tag to accumulate the fee. In positive prorating, select whether or not to prorate the amount. For postpaid subscribers, a tag determines how to align the recurring charge with the bill cycle period. The recurring charge is either invoiced in advance, in the previous billing period, or in arrear, in the current billing period. Positive prorating for activation can only be aligned with the current bill cycle period. The amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context.global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes.
Prolong unit validity by halt period	Select one or more product buckets for which to prolong the unit validity. Select if the extension should be to end of day on the new valid to date. The parameter value is optional to define.
Prolong unit validity to end of recurrence period	Select one or more product buckets for which to prolong the unit validity. The parameter value is optional to define.
Remove from Product Bucket	Select a product bucket and enter the value to remove. Select whether or not to prorate the amount. For monetary transactions, the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting



Parameter Value	Description
	category or tax class for financial and reporting purposes.
Send End-user Notification	Select the notification to send. A description of the selected notification is displayed. The available values and descriptions are fetched from NTF. Select whether to send the notification immediately or at a defined time offset relative to the recurrence event. The parameter value is optional to define.
Send External Notification	Select the action type and the product offering the action applies to. The action type determines the type of action to trigger. The available values are fetched from global data item. Select whether to send the notification immediately or at a defined time offset relative to the recurrence event. The parameter value is optional to define.
Set Product Bucket Value	Select a product bucket and enter the value to set the bucket to. In positive and negative prorating, select whether or not to prorate the amount. The parameter value is optional to define.
Set Characteristic Value	Select a value from the list of defined values for the characteristic. Only single values are available to be set. The parameter value is optional to define.

Service Triggered

Connect a service triggered strategy created in the product specification and configure its values.

At least one service triggered configuration must be included for each service triggered event connected to the product offering.

A service triggered configuration includes actions from the service triggered strategy created on the product specification. For dimensions, use values created on product specification level, or create new ones in the configuration tab. Create values for actions in the configuration tab. At least one statement must be configured.

Service Triggered, once configured, is evaluated from top statement, to bottom statement.

The following options can be selected when setting values for the conditions. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 14 Service Triggered Condition Values

Parameter Value	Description
Available Units	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit.



Parameter Value	Description
	Note: If a referenced characteristic has no defined value, the condition is evaluated to false.
Bucket Balance	Select the product bucket or bucket tag to evaluate, or billing account bucket is displayed. Define the upper and lower limit. Enter a numerical value or reference to a value in a characteristic. The numerical value can be included in the tiered interval. The tiered interval can be selected to have no upper or lower limit. Note: If a referenced characteristic has no defined value, the condition is evaluated to false.
Replenishment Triggered By	Select a product bucket specification. If the product bucket triggering the event is the same as the selected one, the condition evaluates as true.
User Defined Parameter Values	Define the value to compare the common characteristic against. The selected common characteristic can be compared against one of the following values: — A predefined single value in the same characteristic. — A custom value within a range defined in the same characteristic. — A second common characteristic. Available options depend on the characteristics defined in the characteristic set specification.

The following options can be selected when setting values for the actions. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 15 Service Triggered Action Values

Parameter Value	Description
Accumulate Activation Events	Select the product bucket or bucket tag to accumulate the number of activation events. The parameter value is optional to define.
Accumulate Replenishment Events	Select the product bucket or bucket tag to accumulate the number of replenishment events. The parameter value is optional to define.
Add to Product Bucket Note: It is only possible to add units to a product bucket of type Account with multiple value containers.	Select a product bucket and enter the value to add to it. Note: Units cannot be added to product buckets of type account with a single value container. Set the validity of the added units. If a start date is not defined, validity starts at trigger time of the service triggered strategy. If a duration or end date is not defined, the validity is unlimited. For monetary transactions, the amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the



Parameter Value	Description
	price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes.
Deduct from Bucket	Select the product bucket or bucket tag to deduct from, or billing account bucket is displayed. More than one bucket tag can be added. The bucket tags in the list are evaluated from top to bottom. Drag the product buckets or bucket tags to reorder them in the list. For a bucket tag, only referenced buckets defined with monetary units are processed during evaluation. Enter the monetary fee to deduct. Where applicable, select the product bucket or bucket tag to accumulate the fee. The amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a data item, defined in billing & finance type global data item, for grouping fees related to the same type of usage. For example, an invoice can show a separate subtotal for charges incurred while roaming. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Net is only available in a postpaid payment context. Where applicable, select the GL code, taxation class, or both, to tag the monetary transaction with. The available values are fetched from global data item. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes.
Send End-user Notification	Select the notification to send. A description of the selected notification is displayed. The available values and descriptions are fetched from NTF. The parameter value is optional to define.
Send External Notification	Select the action type and a product offering the action applies to. The action type determines the type of action to trigger. The available values are fetched from global data item. The parameter value is optional to define.

Partner Settlement

Connect a partner settlement strategy created in the product specification and configure its values.

At least one price configuration must be included for each partner settlement event connected to the product offering.

For each partner settlement event it is possible to define one default price configuration. The default price configuration is the one used if no other price configuration is selected for the product. If a default price configuration has been selected in a previous active version, then one must also be selected in the newer version. In a newer version the price configuration can be changed but the payment context must either correspond to the payment context in the previous version, or be generic.



A partner settlement price configuration includes actions from the partner settlement strategy created on the product specification. For dimensions, use values created on product specification level, or create new ones in the library. Create values for actions directly on the canvas. At least one statement must be configured.

Partner settlement, once configured, is evaluated from top statement, to bottom statement. The statements can be reordered by dragging a statement to the desired position.

The following options can be selected when setting values for the dimensions.

Table 16 Partner Settlement Condition Values

Parameter Value	Description
Price Type	Select the price type to evaluate and optionally a price subtype. Available values for price type are fetched from common business data.

The following options can be selected when setting values for the actions. Depending on the strategy created in the product specification, different options from the table below are displayed.

Table 17 Partner Settlement Action Values

Parameter Value	Description
Apply Absolute Settlement	Enter a value for the fixed amount to be added to the settlement account bucket. The amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a category, defined in common business data, for grouping fees related to the same type of usage. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Select the GL code and taxation class to tag the monetary transaction with. The available values are fetched from common business data. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.
Apply Revenue Share	Enter a value to add to, or deduct from, the settlement account bucket per earned revenue. The amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a category, defined in common business data, for grouping fees related to the same type of usage. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Select the GL code and taxation class to tag the monetary transaction with. The available values are fetched from common business data. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.
Reclaim Absolute Settlement	Enter a value for the fixed amount to be removed from the settlement account bucket. The amount type and pricing action type, to tag the monetary transaction with, is displayed or selected. Pricing action type is a category, defined in common



Parameter Value	Description
	business data, for grouping fees related to the same type of usage. The amount type is either net or gross. Net is the product price excluding taxes and gross is the price with taxes included. Select the GL code and taxation class to tag the monetary transaction with. The available values are fetched from common business data. The tags enable post-processing systems to group transactions based on their general accounting category or tax class for financial and reporting purposes. The parameter value is optional to define.



3 Product Offering Templates

The Product Offering Templates (POTs) are used by an External Catalog to create Product Offerings. These POTs are exposed via an external API (BSSF Product Catalog Integration REST), and through this API, an External Catalog Manager (ECM) can read all POTs or any specific POT and create Product Offerings using them. During creating the PO the external system uses certain action attributes which are listed and can be referred from External Catalog Integration.

The Product Offering Templates are created using the **Product Offer Configuration** UI application in the **Business Application Portal** menu.

The POTs are created for the following PO types:

- Base
- Discount
- Bundle
- Exchange
- Loan

Note: The POTs are not created for Partner Settlement type.

The POT version created is always valid from beginning of time to end of time.

3.1 Product Offering Templates Configuration

The product offering templates are used to create new product offerings. These offerings use the provided POT as input. Creating a product offering is only possible when the POT is in active state. If an external client sends a request to create a PO using a suspended POT, the request gets rejected with an validation. In such cases, ECM must update the POT state within the ECM.

The ECM is responsible to verify that every product in the external catalog has a distinct and recognizable external ID, to prevent any duplication among different products. RMCA validates the uniqueness of the external ID within a single system and across product offerings. When getting a PO request from external client, RMCA validates the external ID and detects duplicates (such as the same external ID at the product offering level), it rejects the creation of the PO and sends an error message to external systems.

The PO created through the API will be in unlocked state.

The **Product Offering Templates** list view presents all existing POTs and an option to create a new one.



To **Create**, select the product offering template type to create: base, discount, bundle, exchange and loan. Select a product specification from the context menu. The context menu displays product specifications in active state.

Use the **Filter** to search for an existing POT. Refine the search by applying **More Filters**.

Click a template in the list view to edit, delete, or publish a draft, or to view and duplicate a Published template.

In test systems, the options to unpublish, publish, and lock are in active state for POT.

3.1.1 Details Tab in Product Offering Template

Enter a name, description, and external ID. The external ID needs to be unique to address it correctly. The assigned ID is unique and cannot be edited. Toggle **Only use in bundles** option to **Yes** to restrict using the template to bundles only.

Version

In this section, validity period, state, creation date and last modification date of the current template are displayed. A note unique to each template can be added to describe, for example, the reason for creating the template.

Related product specification

The product specifications related to this POT are displayed. Click the eye icon to navigate into the respective **Relations** tab.

3.1.2 Life Cycle Tab in Product Offering Template

A product has a lifecycle during which it transitions through different states, for example from available to active to expired. End-user notifications can be sent at, and at defined offsets prior to, product lifecycle state transitions.

Activation

Product activates on

It specifies whether the active period starts on product creation, on a specified date, or on first traffic event. Activation by first traffic event is only available when the product offering is connected to a sharing specification or an RFSS with provisioning system set to CPM.

For products activated by first traffic event, specify whether the product is available for activation from product creation or from a specified date. The availability period can be set to have no end date, to expire after a specified duration, or to expire on a specific date.



Expiration

Product expires on:

For all activation methods, the product can be set to have no end date, to expire after a specified duration, or to expire on a specified date.

When setting a specified duration or date, the user can choose to extend the validity to the end of the day upon resumption.

End-user Notifications

End-user notifications can be sent at defined offsets prior to the product lifecycle state transitions from one to another state. The following end-user notifications can be defined to be sent:

- Availability expiration reminder notification
- Availability expiration notification
- Activation notification
- Expiration reminder notification
- Expiration notification
- Termination reminder notification
- Termination notification

Note: Product expiration notifications can be defined even if the product offering is configured without an expiration date. This is because the state transition dates can be personalized at instance level.

3.1.3 Relations Tab in Product Offering Template

In the **Relations** tab, the user can connect other product offering templates to the product offering template.

In the **Relations > Product specification** tab, the connected product specifications are listed and can be viewed.

In the **Product Offerings** tab, either select type of relation as **Optional for** or **Activate on expiry**.

If the relation type is selected as **Optional for**, then the related product offering is to be offered as an add-on product to the new product offering template.

If the relation type is selected as **Activate on expiry**, then the new POT will be activated after the trial period expires in the related product offering.



Table 18 Product Offering Templates to Product Offering Template Relations

Area	Description
Library	The type of relation is selected either as Optional for or Activate on expiry. The library lists existing entities that can be connected. Use the Filter to search for an entity. Narrow the search by selecting different variables. Click the Add icon on an existing entity to add it to the Product offerings tab. Click the expandable arrow icon at the bottom of the library sidebar to expand the library and see more detailed information on the entities.
Canvas	The canvas defines the relation. Select, or drag and drop, entities from the library to the canvas to relate them to the product offering template. After a product offering template has been added for a relation type, use context menu to Change type of relation, Go to specification and Remove relation. Relation Graph This tab gives an overview of the product offering template and its connected entities. A relation graph visualizes how the entities relate to each other. The following actions are possible: — Use the filter above the graph to search an entity by its name, type, and resource context where applicable. Click Remove filter to reset the filter. — Click Plus/Minus icon at the top of the graph-field to expand/collapse the nodes along a vertical line. — Click the Plus icon on the main node to view the connected nodes. — Click the Minus icon on the main node to close the branch. — Hover over or click a node for a focused view of the node and those connected to it. A click also displays the relation types. — Click a node to view detailed information about the entity in the details view to the left. Use the context menu in the details view to go to the configuration page of the entity. — Click on the graph to drag it around in the window. — Use the zoom slider at the bottom-right corner of the relation graph to adjust zoom.

3.1.4

Characteristics Tab in Product Offering Templates

Characteristics can be inherited from the connected product specification.

Clicking the **Inherit** option present in the top right corner opens a list view with all the characteristics inherited by the product specification. The **Inherit** option is enabled only if there are any inherited characteristics available for the connected product specification.



Select an inherited characteristic displayed in the list view and click **Inherit**. The characteristic will be inherited to the characteristics of the product offering template.

Clicking on the inherited characteristic displayed in the list view opens the **Edit characteristic** dialog. The dialog consists of the following three sections:

— **Details**

Edit the characteristic details.

— **Settings**

If customization is enabled, edit the level of personalization at instance level. Edit the number of required and allowed values.

— **Values**

Edit the characteristic values. Select which individual values or ranges to use from the original characteristic. New single values can be added if a value range is defined in the original characteristic.

3.1.5

Business Strategies Tab in Product Offering Templates

Business strategies for the POT are defined using this tab. Select from a list of existing business strategies, and create price configuration for the same.

Table 19 Business Strategies Tab

Area	Description
Library	It lists the available business strategies. Click on any business strategy to view its details in the canvas.
Canvas	The canvas defines the connection. It consists of a number of tabs, an Overview and one tab for each strategy type. Navigate between the strategy tabs to add business strategies. Based on the selected type of product specification, the following different strategy tabs will be available for configuration. — Service usage — Refill — Recurrences — One time charge — Aggregated charges — Promotions — Life cycle — Service triggered — Scenario triggered



Area	Description
	For each available strategy tab, at least one price configuration must be configured. Click Create price configuration present at the bottom of the tab or click Create present in the right corner of Price Configurations. Enter or select the relevant details. Select Validate for any errors, and click Done to save the added configuration.

For detailed description on each business strategy type, see [Business Strategies Tab in Product Offerings](#) on page 26.

3.2 Configure Product Offering Templates

Create and manage product offering templates by using the Business Application Portal GUI.

For more information on tasks that can be performed, see the following sections:

- Create a product offering template, see [Create a Product Offering Template](#) on page 52 section.
- View a product offering template, see [View a Product Offering Template](#) on page 57 section.
- Modify a product offering template, see [Modify a Product Offering Template](#) on page 58 section.
- To publish a product offering template, see [Publish a Product Offering Template](#) on page 57 section.
- Duplicate a product offering template, see [Duplicate a Product Offering Template](#) on page 58 section.
- Delete a product offering template, see [Delete a Product Offering Template](#) on page 59 section.

3.2.1 Create a Product Offering Template

A product offering template is created through the **Product Offering Templates** tab which is accessed through the **Product Offer Configuration** UI application in the **Business Application Portal** menu.

Prerequisites

- A product specification.

Steps

1. In the **Product Offer Configuration** UI application, click **Product Offering Templates**.



2. In the **Product Offering Templates** list view, click **Create** in the top right-hand corner.
3. In the **Create product offering template** dialog, do the following steps:
 - a. Select the type of offering under **Details**.
 - b. Click **Next**.
 - c. Select a product specification to create the product offering template from.
 - d. Click **Create**.

Result: The **Details** view of a new product offering template is displayed, showing the sections and fields that create a product offering template.

Note: Depending on the chosen type and the connected product specification, different relationships and business strategies are available for configuration.

Details

4. In the **Details** tab, do the following steps:
 - a. Enter a **Name** for the product offering template.
 - b. Enter a **Description** for the product offering template.
 - c. Enter an **External ID** for the product offering.
 - d. Select the toggle button to **Yes** if this product offering template is to be **Only use in bundles**.

Result: The details of the product offering template are defined.

Life Cycle

5. In the **Life Cycle** tab, do the following steps:
 - a. Under **Activation**, define when the product activates.
 - b. If the product is activated by the **First traffic event**, define when the activation availability period starts and ends.
 - c. Under **Expiration**, define when the product expires.
 - d. If either **Duration** or **Date** are defined, then the user can select **Extend validity to end of day on resumption** to extend the validity to the end of day on resumption.
 - e. Different types of End-user notifications can be defined for the POT. To define, select and drag a notification to send.

Note: Only one end-user notification can be added for each type.



Result: The life cycle for the POT is defined and displayed under the **Life Cycle** tab.

Loan Due Date

Note: If the POT type is chosen as **Loan**, the **Loan Due Date** tab is enabled.

6. In the **Loan Due date** tab, perform the following steps:
 - a. Select the default **Due Date** for the loan product offering template.
 - b. Define the **End-user notifications**.
 - Under **Warning**, select and drag a warning notification from the library, and define the period of the time before due date the warning is to be sent.
 - Select and drag a notification on the **Due date**.
 - c. Define the **External system notifications**.
 - Select and drag a notification on the **Due date**.
 - Select and drag a notification on the **Product deactivation**.

Result: The Loan Due Date of the loan product offering template is defined.

Relations

7. In the **Relations** tab, do the following steps:
 - a. In the **Product offerings** tab, select the type of relation and click **Next**.
 - b. Select and drag one or more product offerings to connect with the selected relation type.
 - c. Click **Save**.

Result: One or more product offerings are selected and shown under the **Product offerings** tab.

Characteristics

8. In the **Characteristics** tab, inherited characteristics can be customized. To do, perform the following steps:
 - a. Click **Inherit** to inherit a characteristic to the product offering template.

Result: All the inherited characteristics are listed.
 - b. Select a characteristic to customize.
 - c. Click **Inherit**.



Result: The selected characteristic is inherited to the Characteristics of the product offering template.

- d. Click on the inherited characteristic.

Result: The **Edit characteristic** dialog is displayed. The dialog consists of three sections: **Details**, **Settings** and **Values**

- e. In the **Edit characteristic > Details**, edit the **Name** and **Description** for the characteristic, if required.
- f. Edit the auto-generated **External ID** of the characteristic.
- g. Click **Next**. The **Settings** is enabled.
- h. If **Enable customization** is set to **Yes** for the characteristic, choose the **Personalization on instance level** and if required, select the **Allow values to be transferred** check box to enable values to be transferred from the source to the target product instance at product exchange.
- i. Under **Cardinality**, enter the **Number of required values** and **Number of allowed values**.
- j. Click **Next**. The **Values** section is enabled.
- k. Define any **Single values** by entering **Name**, **Value**, and **Unit**.
- l. Edit or restore the **External ID**. Select if the value is to be used as default.
- m. Define any **Value range** by entering **Name**, **Min value**, **Max value**, and **Unit**. Edit or restore the **External ID** or select **Use full range** of the selected value type.
- n. Click **Done**.

Result: The characteristic is customized and shown under the **Characteristics** tab.

Business Strategies

9. In the **Business Strategies** tab, do the following steps:
 - a. Select the tab of the type of business strategy to edit: Service Usage, Refill, Recurrences, One time charge, Aggregated charges, Promotions, Life cycle, Service triggered, Scenario triggered.

Note: Depending on the selected business strategy, different options are available for configuration.
 - b. Select an existing strategy from library.
 - c. Define **Default Price Configurations** for each price group.



- d. To create a new price configuration, click **Create price configuration** present at the bottom of the page or click **Create** present in the right corner of Price Configurations.
- e. In the dialog that appears, enter **Name**, **External ID** and select **Payment context**.
- f. Define **Configuration** values depending on each business strategy. For detailed information, see [Business Strategies Tab in Product Offerings](#) on page 26.

Note: The values for **Change product state** which are available for the business strategy types such as Recurrence, Promotions and Scenario Triggered are not available for POT.

The **Activate reward product** available for the Promotions will allow only a single value for configuration.

- g. In the configuration tab, perform the following:
 - i. Click plus icon to define the conditional parameters.
 - ii. Click pencil icon to define the action values.
 - iii. Click the context menu in the right corner of each configuration row to move the rows up and down, insert new rows, Duplicate actions , and auto-generate the rows.
- h. Enter an **External ID** for the price configuration.
 - i. In the **External Id** tab, edit or restore the auto-generated external ID of the applicable row(s), action(s), or attribute(s).
 - j. In the **Personalization** tab, select the applicable row(s) and click **Enable personalization**.
 - k. Click **Validate** to check for any errors. If there are any errors, they will be displayed at the bottom of the page.
 - l. Click **Done**.

Result: A price configuration is created for a business strategy and shown under the respective strategy tab.

- 10. Click **Save** in the top right-hand corner of the product offering template detail view.

Result: By default, a product offering template is saved in draft state and can be edited or deleted. To publish the product offering, click **Publish**.



3.2.2 View a Product Offering Template

Entities are viewed from their respective list views, which are accessed through the UI applications in the **Business Application Portal** dashboard.

Steps

1. In the UI application, click the tab of the type of entity to view.

Result: The entity list view is displayed, showing a list of existing entities.

2. Click an entity.

Result: The selected entity is displayed.

Note: An entity in **Published** state is viewed as read-only, while an entity in **Draft** state can be modified.

3.2.3 Publish a Product Offering Template

Entities in draft state are published from their respective list views, which are accessed through the UI applications in the **Business Application Portal** dashboard.

Note: Publishing an entity also publishes all connected entities in draft state.

Steps

1. In the UI application, click the tab of the type of entity to publish.

Result: The entity list view is displayed, showing a list of existing entities.

2. Click an entity that is in **Draft** state.

Result: The selected entity is displayed.

3. Click **Publish** in the top right-hand corner of the page.

Result: A publish confirmation dialog is displayed.

4. Click **Publish**.

Result: The entity is successfully moved to **Published** state and is viewed as read-only.



3.2.4 Modify a Product Offering Template

Entities in draft state are modified from their respective list views, which are accessed through the UI applications in the **Business Application Portal** dashboard.

Note: A user can only modify the **Draft** version of an entity. An entity in **Published** state cannot be modified.

Steps

1. In the **Product Offer Configuration** UI application, click **Product Offering Templates**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. Click an entity that is in **Draft** state.

Result: The selected entity is displayed.

3. Modify any of the editable fields of the entity.
4. Click **Save**.

Result: The modifications of the entity are saved.

In some views, it is possible to revert changes made to an entity, as long as the changes are not saved.

To revert changes, click **Revert Changes**. In the confirmation dialog that appears, click **Revert Changes**. The modifications of the entity are discarded.

3.2.5 Duplicate a Product Offering Template

Entities in published state are duplicated from their respective list views, which are accessed through the UI applications in the **Business Application Portal** dashboard.

Steps

1. In the UI application, click the tab of the type of entity to duplicate.

Result: The entity list view is displayed, showing a list of existing entities.

2. Click an entity that is in **Published** state.

Result: The selected entity is displayed as read only.

3. Click **Duplicate** in the top right-hand corner of the page.



Result: The entity is duplicated. The duplicated entity is displayed in **Draft** state and can be modified.

4. Click **Save** to save the duplicate.

Result: By default, the duplicate is successfully saved in **Draft** state and can be modified.

Note: In some views, it is possible to discard duplication of an entity, as long as it is not saved. To discard, click **Discard**. In the confirmation dialog that appears, click **Yes**.

The duplicated entity is discarded.

3.2.6 Delete a Product Offering Template

Product offering templates in published state are suspended from their respective list views, which are accessed through the **Product Offer Configuration UI** application in the **Business Application Portal** menu.

Steps

1. In the **Product Offer Configuration UI** application, click **Product Offering Templates**.

Result: By default, the entities in draft and published state are displayed in the list view.

2. To Delete a product offering template, do any of the following:
 - a. Click the context menu option present at the right side of the **Published** entity row to expand it.
 - Or-
 - b. Click an entity that is in **Published** state to open the **Details** tab view and click the context menu in it.

Result: The context menu list is displayed.

Note: The **Delete template** option is enabled when the product offering template is being used by a tag specification or NPG.

3. Click **Delete template** in the list.

Result: A suspend confirmation dialog is displayed.

4. Select the date on the confirmation dialog display, If required.



Note: If a date is selected to define the grace period, the product offering template will be scheduled to get suspended on the selected date and it will be permanently deleted from the system after the grace period is over.

If a date is not selected and a grace period is not defined, the product offering template gets suspended. The suspended product offering template is grayed out in the list view and displayed as Suspended.

5. Select the confirmation check box, and click **Delete Product offering template**.

Result: If grace period is defined, the product offering template gets scheduled for suspension. Product offering template will get deleted after the grace period is over. A yellow warning icon with the product offering template is displayed in the list view. Hover over the icon to view suspension date.

If grace period is not defined, the product offering template gets suspended and a message stating that the visibility of the product offering template for other systems has been suspended is displayed. The suspended entity is grayed out in the entity list view, and displayed as Suspended.



4 Product Priorities

4.1 Recurrence Priority Configuration

Recurrences apply to the payments that are scheduled to be executed.

Recurrence priority list defines the order of recurrent jobs to be executed for the available products. Priority can be defined both among different product offerings, and within each product offering. Recurrence priority list can be configured only for the regular recurrences. Only one recurrence priority list can be configured.

A blank recurrence priority list is pre-configured where the product offerings can be prioritized.

Click on the recurrence priority list displayed in the list view to edit, delete, or publish the draft, or to view or create a new version of the published priority list.

In a published entity with a newer version in draft state, use the drop-down next to the specification title to switch between the versions. Different entities permit different changes in new versions.

Note: In a test system, unpublishing the entities listed in a priority list, removes them from the list and creates a new version of the priority list. When an entity is unpublished or deleted, the priority list in which the entity is listed must be reconfigured.

4.1.1 Recurrence Priority List in Recurrence Priority

Define the priority between recurrences.

Published product offerings with recurrences, and recurrences within each published product offering, are prioritized relative to each other. This determines the execution order of recurrent jobs triggered on the same day and time. A higher position in the list corresponds to a higher priority. The following actions are possible:

- Use the search on the left-hand side of the page to find a specific product offering or recurrence within the list.
- Move a row up and down in the list by using the arrows and context menu on the right-hand side of the row.
 - Rows in nested lists can only be moved to other positions inside same nested list.

Note: Recurrences are always prioritized higher than product offerings inside nested lists.



- Rows outside nested lists cannot be moved to positions inside any nested list.

A recently published product offering with recurrence is automatically added to the bottom of the list with the lowest priority.

The recurrence priority list hierarchy is defined as follows:

Table 20 Recurrence Priority List Hierarchy

Level	Contains
Top level	— Product offerings with recurrence but without use restricted to bundles only — Bundle product offerings with at least one child with recurrence
Child level	First: — Recurrence Second: — Product offerings with recurrence — Bundle product offerings with at least one child with recurrence

Version

Current entity validity period, state, creation date, and last modification date are displayed. A note unique to each version of the specification can be added to describe, for example, the reason for creating the version.

4.2 Service Usage Priority Configuration

A service can be used by multiple product offerings. The service usage priority list displays the priority between all product offerings that use the same service. A priority sorting rule defines how the list is sorted.

The service usage list view presents all existing service usage priorities. One priority list exist for each service that has a connection to at least one active product offering.

Use the filter to search for a service usage priority list. Sort and group the lists by selecting different variables.

Click on a service usage priority list displayed in the list view to edit, delete, or publish a draft, or to view or create a new version of a published priority list.

Note: In a test system, deactivation of the entities listed in a priority list removes them from the list and replaces the priority list with a new version. When an entity is deactivated or deleted, the priority list in which the entity is listed must be reconfigured.



4.2.1 Service Usage Priority List in Service Usage Priority

The name and subtype of the service are displayed.

Equal Priority Sorting Rule

Equal priority sorting rule defines how products with the same priority are sorted.

Table 21 Equal Priority Sorting Rule

Equal Priority Sorting Rule	Description
Expiry Only	Sorts products by the following order: 1. Expiry date - earlier expiry date has higher priority If there is no expiry date defined for a product, validity is considered as end of time. 2. Lexical order - lexical order of the product IDs
Due Date First	Sorts products by the following order: 1. Due date - earlier due date has higher priority 2. Expiry date - earlier expiry date has higher priority If there is no expiry date defined for a product, validity is considered as end of time. 3. Activation date - earlier activation date has higher priority 4. Lexical order - lexical order of the product IDs If there are products both with and without due date, those with due date are prioritized over the others.

Product Priority List

Define the priority between product offerings.

Active product offerings that use the service can be prioritized relative to each other. This determines in which order the products are used for the service. A higher position in the list corresponds to a higher priority. The following actions are possible:

- Use the search on the left-hand side of the page to find a specific product offering or priority group within the list.
- Move a row up and down in the list by using the arrows and context menu on the right-hand side of the row.
 - Rows in nested lists can only be moved to other positions inside same nested list.
 - Rows outside nested lists cannot be moved to positions inside any nested list.
- Select multiple rows to group product offerings that are to have the same priority. The sorting rule defines how product offerings with the same priority are sorted.



- Select one or more priority groups to ungroup product offerings that are not to have the same priority. To move a single product offering out of a group, use the move to position option and define a new position for the product offering.

A recently activated product offering is automatically added to the bottom of the list with the lowest priority.

The service usage priority list hierarchy is defined as follows:

Table 22 Service Usage Priority List Hierarchy

Level	Contains
Top level	— Base product offerings
Child level	— Discount product offerings

Only product offerings with a direct relation to the RFSS through a product specification and CFSS are included in the priority list (Product Offering>Product Specification>CFSS>RFSS).

Version

Current entity validity period, state, creation date, and last modification date are displayed. A note unique to each version of the specification can be added to describe, for example, the reason for creating the version.

4.3 Aggregated Charges Priority Configuration

Aggregated charges apply to service usage consumption accumulated to a periodic counter during a bill period.

Aggregated charges priority list defines in which order aggregated charges is to be deducted for the available products. Priority can be defined among different product offerings, and within each product offering. Aggregated charges priority list can be configured only for base product offerings with aggregated charges. Only one aggregated charge priority list can be configured.

A blank aggregated charges priority list is pre-configured where the product offerings can be prioritized.

Click the aggregated charges priority list displayed in the list view to edit, delete, or publish the draft, or to view or create a new version of the published priority list.

In a published entity with a newer version in draft state, use the drop-down next to the specification title to switch between the versions. Different entities permit different changes in new versions.



Note: In a test system, unpublishing the entities listed in a priority list, removes them from the list and creates a new version of the priority list. When an entity is unpublished or deleted, the priority list in which the entity is listed must be reconfigured.

4.3.1 Aggregated Charges Priority List in Aggregated Charges

Define the priority between product offerings and their periodic counter product buckets.

Published product offerings with aggregated charges, and periodic counters within each published product offering, can be prioritized relative to each other. This determines in which order aggregated charges are deducted for service usage consumption. A higher position in the list corresponds to a higher priority. The following actions are possible:

- Use the search on the left-hand side of the page to find a specific product offering or periodic counter within the list.
- Move a row up and down in the list by using the arrows and context menu on the right-hand side of the row.
 - Rows in nested lists can only be moved to other positions inside same nested list.
 - Rows outside nested lists cannot be moved to positions inside any nested list.

A recently published product offering with aggregated charges is automatically added to the bottom of the list with the lowest priority.

Table 23 Aggregated Charges Priority List Hierarchy

Level	Contains
Top level	— Bundle product offerings — Product offerings
Child level	— Product bucket specification of type periodic counter Only periodic counters bound to an aggregated charges strategy are listed.

Version

Current entity validity period, state, creation date, and last modification date are displayed. A note unique to each version of the specification can be added to describe, for example, the reason for creating the version.



5 Common UI behavior

5.1 Price Configuration Window in Product Domain

Create values for the following strategies configured in the product offering by using the price configuration window:

- Service Usage
- Refill
- One Time Charge
- Discount on One Time Charge
- Promotions
- Life Cycle
- Recurrences
- Service Triggered
- Product Exchange
- Partner Settlement

Condition parameters, and action parameters, are building blocks used to define the price configuration. The building blocks contain parameter values.

Condition parameters and action values are created in the configuration tab.

To open a price configuration window, click **Create price configuration** in the business strategy tab. Click **Create** to add a new price configuration to a strategy where at least one price configuration already exists.

The following figure shows the price configuration window.

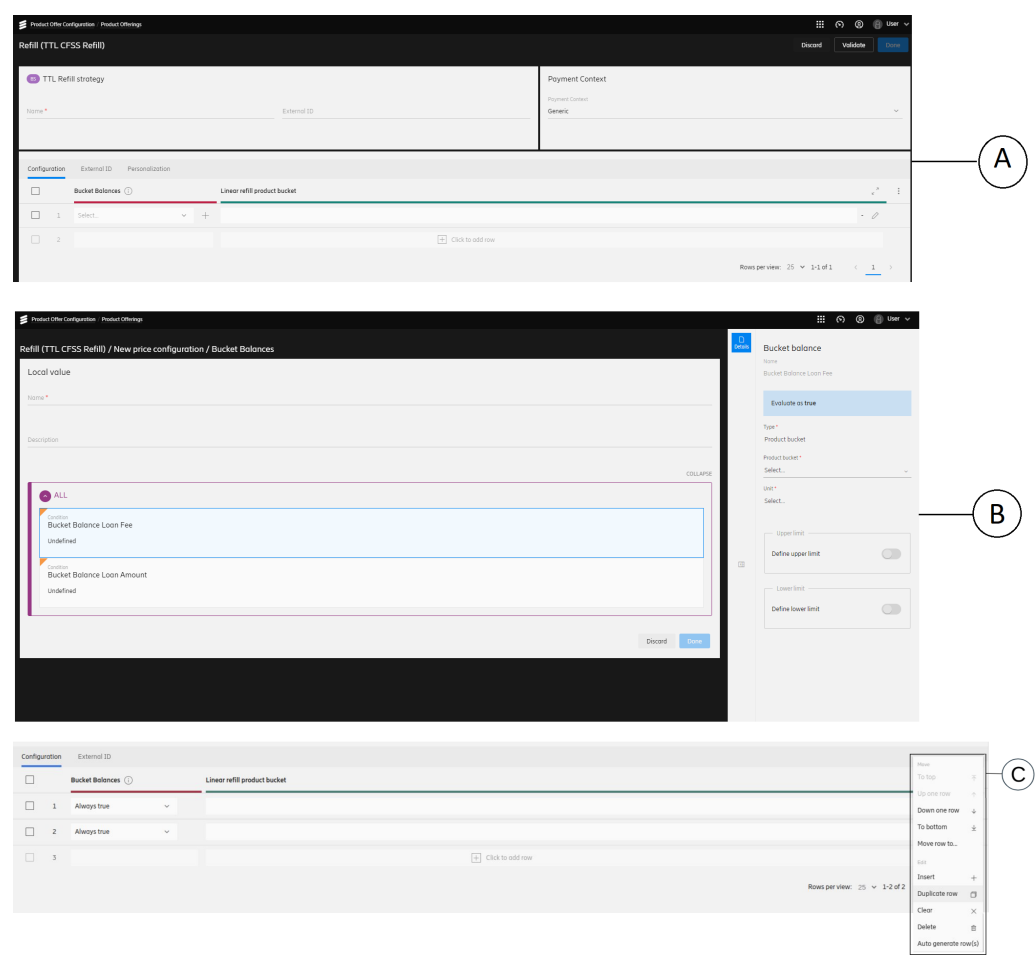


Figure 1 Overview of the Price configuration window

Key

A Price configuration window

B Local value pane

C Context menu

The following table describes the common behavior of the price configuration window:

Table 24 Common Behaviors of the Price Configuration Window

	Description	Action
External ID	The ID used by external systems to identify the price configuration.	Enter an external ID.
Payment Context	The payment context of the configuration is set to, prepaid, postpaid, or generic depending on the payment method to be used.	Select payment context in the drop-down.



	Description	Action
Enable Personalization	The ability to personalize values of instantiated products is defined.	Select the check box to enable personalization in the personalization tab.
Statement	A statement is a fully configured row on the canvas. A statement consist of one action parameter and where a applicable one or more condition parameters. Statements are evaluated from top to bottom.	To create a statement, define an action parameter and where applicable its corresponding condition parameters.
Local value pane	In the local value pane, condition parameters are created. The local value pane displays what condition parameters are available for each condition.	Click plus icon under the configuration tab to create a new condition parameter.
Validate	Validates the values configured in the price configuration window. If any missing or faulty values are discovered, it shows the summary at the bottom of the page shows the number of errors and warnings.	Click Validate to validate the values configured in the price configuration window.
Context Menu	Context menu in the configuration tab, allows the user to move or edit the rows in batch.	To rearrange the rows, click options under the move section. Under the Edit section, click Insert to add a new statement. Click Duplicate action (s) to copy the values to another row. Click Auto generate row (s) and select conditions to auto generate the rows with all possible combinations in batch. Existing combinations are not generated again. Click Delete to delete a row. Click Clear to delete all values in a row.
Condition Parameter	Condition parameters are building blocks created, under the condition name previously created in the connected strategy.	To create a condition parameter, click plus icon or expand button in the configuration tab.
Global Condition Parameters	A global condition parameter exists globally and can be reused on other product offerings. If a global condition parameter has a reference to an entity that requires a relation to the product offering but the relation does not exist, the condition parameter is invalid. A condition parameter not set to global, is local. A local condition parameter exists on a specific product offering and cannot be reused.	For global conditions, select to save the condition parameter as global or local. The selection cannot be changed after the created condition parameter is saved. To create a condition parameter, open the value panel, and addLocal conditions saves the condition parameter as local, and no choice is presented. Deleting or not saving a configuration does not delete saved global conditions parameters.
Always True	The always true condition parameter, always triggers the related action. This can be used independently, or combined with other condition parameters.	The always true condition parameter is located under the configuration tab
Action Parameter	Action parameters are defined by one, or more, parameter values, and are executed when all its given conditions are evaluated to true. Action parameters are created using pencil icon or expand button in the configuration tab . If no condition is defined, the action is always evaluated to true.	To create an action parameter, click pencil icon or expand button in the configuration tab.



	Description	Action
Parameter Values	Values for condition or action parameters, that are used in the evaluation of the configuration. Some parameter values are optional to apply, but an action parameter must always have at least one parameter value defined. Note: When using characteristics or service specific parameters in conditions, value ranges can only be compared with single values, and not with other value ranges. A system defined condition parameter without a specified value is evaluated as always true, with the exception of bucket balance, available units and community membership. For bucket balance and available units to achieve the always true behavior, Unit type has to be selected, as well as No upper limit and No lower limit. Community membership has no always true behavior, but must have specified values. A service specific or a user defined condition parameter must also have specified values.	To create a value parameter, click plus icon in the configuration tab

5.2 Conditions and Actions Matrix in Product Domain

The following matrix shows which conditions apply, for which business strategies:

Note: Not all strategies are available in all domains.

Table 25 Condition and Strategy Matrix

C o n d i t i o n	Strategy																										
	S e r v i c e U s a g e	O n e T i m e C h a r g e	D i s c o u n t o n O n e T i m e C h a r g e	A g g r e g a t e d C h a r g e s	R e f i l l	R e c u r r e n c e s	S e r v i c e T r i g g e r e d	P r o m o t i o n	L i f e C y c l e	S c e n a r i o T r i g g e r e d	P r o d u c t E x c h a n g e	P a r t n e r S e t t l e m e n t			N e t w o r k A u t h o r i z a t i o n	F i l t e r i n g	S c r e e n i n g	T h r e s h o l d s	O f f s e t	C o n s u m p t i o n P e r m i s s i o n	R e p l e n i s h m e n t P e r m i s s i o n	A c c u m u l a t i o n P e r m i s s i o n	S h a r i n g U s a g e	S t a t u s R e p o r t i n g	E n r i c h m e n t s		
A v a	X	X		X	X	X	X	X	X	X																	



Condition	Strategy																								
	Service Usage	One Time Charge	Discount One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offset	Consumption Permission	Replenishment Permission	Accumulation Permission	Sharing Usage	Status Reporting	Enrichments
ilable Units																									
Bucket Balance	X	X	X	X	X	X	X	X	X	X										X					
Bucket Validity										X															
Characteristics Condition	X	X	X	X	X	X	X	X	X	X	X				X	X	X	X		X	X	X	X	X	X

[illegible]



Condition	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offset	Consumption Permission
thus																				
NESP parameters																				X
Previous products state						X														
Pricetype												X							X	X
Reason for						X														

C o n d i t i o n	Strategy																										
	S e r v i c e U s a g e	O n e T i m e C h a r g e	D i s c o u n t o n O n e T i m e C h a r g e	A g g r e g a t e d C h a r g e s	R e f i l l	R e c u r r e n c e s	S e r v i c e T r i g g e r e d	P r o m o t i o n	L i f e C y c l e	S c e n a r i o T r i g g e r e d	P r o d u c t E x c h a n g e	P a r t n e r S e t t l e m e n t			N e t w o r k A u t h o r i z a t i o n	F i l t e r i n g	S c r e e n i n g	T h r e s h o l d s	O f f s e t	C o n s u m p t i o n P e r m i s s i o n	R e p l e n i s h m e n t P e r m i s s i o n	A c c u m u l a t i o n P e r m i s s i o n	S h a r i n g U s a g e	S t a t u s R e p o r t i n g	E n r i c h m e n t s		
r p r o d u c t s t a t e c h a n g e																											
R e a s o n f o r p r o d u c t h a i t					X				X																		
R e a s o n f o r p r o d u c t					X				X																		



Condition	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offset	Consumption Permission
Resumption																				
Replenishment Triggered by						X														
Scenario							X													
Service Specific para	X			X					X						X	X	X			

C o n d i t i o n	Strategy																										
	S e r v i c e U s a g e	O n e T i m e C h a r g e	D i s c o u n t o n O n e T i m e C h a r g e	A g g r e g a t e d C h a r g e s	R e f i l l	R e c u r r e n c e s	S e r v i c e T r i g g e r e d	P r o m o t i o n	L i f e C y c l e	S c e n a r i o T r i g g e r e d	P r o d u c t E x c h a n g e	P a r t n e r S e t t l e m e n t			N e t w o r k A u t h o r i z a t i o n	F i l t e r i n g	S c r e e n i n g	T h r e s h o l d s	O f f s e t	C o n s u m p t i o n P e r m i s s i o n	R e p l e n i s h m e n t P e r m i s s i o n	A c c u m u l a t i o n P e r m i s s i o n	S h a r i n g U s a g e	S t a t u s R e p o r t i n g	E n r i c h m e n t s		
m e t e r s																											
S t a t e b e f o r e p r o d u c t h a l t						X			X																		
T e s t R a t i n g															X												
T h r e s h o l d																			X								
T i e r e d R	X			X																							

Condition	Strategy																										
	ServiceUsage	OneTimeCharge	DiscountOneTimeCharge	AggregatedCharges	Refill	Recurrents	ServiceTriggered	Promotion	LifeCycle	ScenarioTriggered	ProductExchange	PartnerSettlement				NetworkAuthorization	Filtering	Screening	Thresholds		Offset	ConsumptionPermission	ReplenishmentPermission	AccumulationPermission	SharingUsage	StatusReporting	Enrichments
ating																											
TimeandDayofWeek	X															X											
Timeofday	X															X					X	X	X	X	X	X	

The following matrix shows which actions apply, for which business strategies:

Table 26 Action and Strategy Matrix

Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Lifecycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
Accumulate Activation Events							X													
Accumulate Consumed Units	X			X																
Accu								X												



Action	Strategy																			
	Service Usage	One Time Charge	Discount On One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
Multitask Promotion Events																				
Accumulate One Time Charge Events	X																			
Accum					X															

Action	Strategy																	
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Lifecycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds
Calculate Refill Events																		
Accumulate Refill Value					X													
Accumulate Recurrence Eve						X												



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
nts																				
Accumulate Recurrence Resumption Events					X															
Accumulate product									X											

Action	Strategy																	
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Lifecycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds
Resumption Events																		
Accumulate Replenishment Events							X											
Accumulates										X								



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrents	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
Scenario Events																				
Activate Reward Product								X												
Add to Billing Account Buck					X															

Action	Strategy																	
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Lifecycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds
Offset																		
Consumption Permission																		
Replenishment Permission																		
Accumulation Permission																		
Sharing Usage																		
Status Reporting																		
Enrichments																		
et																		
Add to Product Bucket	X				X	X	X	X										
Align All Recurrences											X							
Align Recurrences by T											X							



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
ag																				
Align Recurrences with Account Units											X									
Apply Absolute Discount	X			X																
App												X								

[illegible]



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
re														is able to configure this action if Tier rating is added as condition.						
Change p					X		X		X											

[illegible]



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
duct from Bucket																				
Deduct Line ar Price	X																			
Deduct Relative Fee				X																
Deduct Se	X																			

[illegible]



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrents	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
ng Bucket Validity																				
Prolong Contract Lifecycle					X															
Prolong unit validity by n						X			X											

[illegible]



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
Inter Settlement																				
Refill Billing Account Bucket Linearily					X															
Refill Billing Account					X															

[illegible]



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
from Product Bucket																				
Replace base price	X																			
Send end-user notification		X			X	X	X	X		X										
Se		X			X	X	X	X		X										

[illegible]



Action	Strategy																			
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission
ment																				
Set Absolute Distributed Discount		X																		
Set Absolute Linear Discount			X																	

Action	Strategy																	
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Lifecycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds
nt																		
Set Characteristic Value					X			X	X									
Set Charging Interval	X																	
Set Permission																		X

[illegible]

[illegible]



Action	Strategy																								
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrents	Service Triggered	Promotion	Life Cycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds	Offsets	Consumption Permission	Replenishment Permission	Accumulation Permission	Sharing Usage	Status Reporting	Enrichments
ue																									
Transfer Account Units											X														
Transfer All Bucket Units											X														
Transfer Coun											X														

Action	Strategy																	
	Service Usage	One Time Charge	Discount on One Time Charge	Aggregated Charges	Refill	Recurrences	Service Triggered	Promotion	Lifecycle	Scenario Triggered	Product Exchange	Partner Settlement			Network Authorization	Filtering	Screening	Thresholds
																		Offsets
																		Consumption Permission
																		Replenishment Permission
																		Accumulation Permission
																		Sharing Usage
																		Status Reporting
																		Enrichments
ter Units																		
Trigger Scenario																	X	X

Note: Do not combine in the same strategy several deduct from product buckets or billing account bucket actions that directly address and also via a tag targets the same product bucket.



6 Batch Configuration

6.1 Batch Configuration in Product Domain

Batch configuration allows multiple entities to be configured as a group and updated in one operation to help reduce overall configuration time. Batch configuration is supported for the following entity specifications:

- Product offering (all types)
- Product specification (all types)
- Physical resource specification

The types of changes that can be applied to a batch of entities are:

Table 27 Batch Configuration Actions

Action	Description
Update identity and state	Publish a draft specification or create a new version of a published one. Edit specification name, description, and external ID fields. In a test system the options to lock and unpublish a published specification are available.

The initial list of candidates for inclusion in a new batch is every existing specification of the chosen type. This can be a very large list. Batch configuration therefore offers both simple and advanced filtering of candidate specifications to help in finding the desired specifications. Advanced filters limit the list of candidate specifications to those with a relation to a specified specification or specifications. For example, a list of product specifications can be filtered to only those with a relation to a specified group of customer facing service specifications.

Batch configuration in a production system is done with production system capabilities and limitations. Batch configuration in a test system is done with test system capabilities and limitations. The batch configuration process is done with relaxed validation to help speed the process of making many updates at once. However, configured changes are only applied if they meet the requirements of the respective system for specifications configured individually in regards to configurable fields, relation types, and valid entity state transitions. Invalid configuration changes are rejected.

To make best use of batch configuration and avoid rejected changes, familiarity with the following is recommended:

- Requirements and limitations that apply for business configuration entities in general and for each specification type specifically.



- Cascading activation behavior, which ensures that publishing a specification also publishes any related lower level entities.
- The applicable versioning behavior for each specification type.
- The extra configuration capabilities available in a test system, including unpublish/deactivation and entity locking.

6.2 Batch Configuration in Configuration Tools UI Application

Use batch configuration to apply changes to multiple entity specifications at the same time. To begin the batch configuration process, first select an entity specification. All specifications in a batch must be of the same type. The specifications available for batch update are:

- Product offering (all types)
- Product specification (all types)
- Physical resource specification

Click **Start** to begin the batch configuration. The process has the following steps:

— Step 1 - Select Specifications

Select specifications to include in the batch. The results list displays all existing entity specifications of the chosen type. Refine the list by using the filter above the header row. Start typing and the filtering begins. Click to apply **More filters** if required. In a test system, locked entities are indicated by a lock in the state column. Hover over the lock to see the lock reason.

The advanced filters library enables filtering of the specifications based on their relation to other entities. To create an advance filter, click **Add** in the library and choose an entity type. In the pop-up, select one or more existing specifications of the chosen type to limit the list of candidate batch specifications to those with relations to these specifications. If multiple advanced filters are created, only candidate specifications that meet the criteria of every filter are returned. Use the context menu on an advanced filter to remove it.

Select one or more specifications in the list and click **Add to batch** to add selected specifications to the batch. Repeat as many times as required with different filter parameters. Specifications already added to the batch are excluded from the results list.

— Step 2 – Review Batch

Review the batch of specifications created in the previous step. Select any unwanted specifications and select **Remove from batch** to remove them. To add more specifications, return to the previous step.



— Step 3 - Select Type of Action

Select the type of changes to apply to the batch:

Table 28 Batch Configuration Actions

Action	Description
Update identity and state	Edit specification names, descriptions, and external IDs. Publish drafts or create new versions of published specifications. In a test system the options to lock and unpublish a published specification are available.

— Step 4 - Configure

Table 29

Action	Description
Update identity and state	Configure the desired changes in the editable fields for each specification in the batch. To change the state of a specification, select its check box or use the context menu in the State change column of the relevant row. When all configuration updates have been made, click Perform Changes to apply the changes to the batch.

Only configured changes that conform to ordinary specification validation rules are applied. After performing changes, a final summary shows which updates were successful. Hover over the tick or cross icon for more information on the updates that succeeded or failed. After performing batch entity configuration changes, use the context menu on an individual entity to open its product specification detail view and perform further changes to that specific entity as required. During batch configuration, the product specification detail view includes a link above the title to go directly back to batch configuration.

Use the arrow buttons to navigate forward and backward between the steps. Entities with several versions are indicated in the list view with the number of versions in parentheses, for example (+1). At any stage, use the context menu in the table title row (above the filter) to export the listed specifications to a CSV file. At the specification selection stage the export includes only the specifications shown on page one and at subsequent stages it includes the whole batch. Use the context menu in the column header row to add or remove columns from the table view.



7 RMCA Business Configuration Rollback

The Business configuration rollback point is automatically created when the RMCA business configuration import is initiated. It helps to restore the configuration to the system version before import. The purpose of the business configuration rollback point is to capture the state before the most recent RMCA business configuration import. It also allows users to preview the list of affected entities for a specific rollback point.

A business configuration rollback point is valid for 7 days (168 hours) from the start of a business configuration import operation. However the rollback point can be configured up to 30 days using the `rollbackWindow` parameter using `mscli` in virtualized deployment or `BEAM CLI` in containerized deployment.

Note: Changing the `rollbackWindow` parameter value would invalidate existing rollback points, preventing the system from restoring the previous configuration. Therefore, update the `rollbackWindow` parameter value only when the current rollback point is no longer needed and before importing a new configuration.

There can only be one business configuration rollback point available at a time. Whenever a new RMCA business configuration import is done, a new rollback point is created and the old rollback point is no longer be available.

When the PO versions are deleted from the GUI and if they have any valid rollback points, it gets invalidated and the rollback cannot be performed.

For more information on tasks that can be performed, see the following sections:

- Preview Rollback Point, see [Preview Rollback Point](#) on page 105.
- Perform Rollback, see [Perform Rollback](#) on page 106.

7.1 Preview Rollback Point

A rollback point review operation is performed from the **Rollback** list view which is accessed through the **Configuration Tools** UI application in the **Business Application Portal** menu.

Steps

1. In the **Configuration Tools** UI application, click **Rollback**.
2. In the **Rollback** list view, do any of the following:
 - a. Choose any rollback point and click on it.

-Or-



- b. Click the context menu option present at the right side of the entity row to expand it.

In the context menu, do either of the following steps:

- i. Click **Review scope** in the context menu.

-Or-

- ii. Click **Rollback**, and in the pop-up that opens, click **Review**.

Result: **Details** and **Summary** related to rollback point, and **Rollback scope** are displayed.

The **Rollback scope** section displays the name, type of specification, rollback action, and last modification date of all specifications that will be affected by the rollback. The following are the two types of rollback actions:

- **Partial removal** - It indicates that the specification will remain after a rollback operation. Only updates related to the specification will be rolled back.
- **Full removal** - It indicates that the specification will be removed after a rollback operation as the creation of the specification itself will be rolled back.

In the rollback scope, use the **Filter** option to search for required specification.

To review the rollback changes for an entity, hover on the right side of it, an eye icon is displayed. Click on the eye icon, to open the **Review changes** dialog.

It displays a table including specification's log entries, details and its rollback effect. The log entry for suspended / unsuspended product offerings , as well deleted obsolete version / recreated version of product offering is always listed at the top of the table.

The suspend / unsuspend product offering and delete / recreate version are also counted in partial removal.

The **Rollback effect** column shows the impact of the PO when it gets rolled back.

7.2 Perform Rollback

A rollback operation is performed from the **Rollback** list view which is accessed through the **Configuration Tools** UI application in the **Business Application Portal** menu.



Steps

1. In the **Configuration Tools** UI application, click **Rollback**.
2. In the **Rollback** list view, click **Rollback** in the top-right hand corner.
3. Select **Yes, I want to perform a rollback**.
4. Click **Rollback to**.

Result: A notification is displayed at the bottom, indicating that the business configuration has been rolled back.